



Partially decentralized control of large-scale variable-refrigerant-flow systems in buildings



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ABSTRACT

We consider the problem of designing a scalable control architecture for large-scale variable-refrigerant-flow (VRF) systems. Using a gray-box modeling approach, and by exploiting the one-way coupling between the fluid dynamics and thermal dynamics, we derive individual linearized models for each class of dynamics. This sufficiently reduces the complexity of the problem so that a scalable analysis is possible. Based on the natural dynamics and coupling of the VRF system, which become apparent through the structure of the derived models, a partially decentralized control architecture is proposed. Communication is limited to one-way sensor information flow from the individual decentralized controllers to a global controller, leading to a simple yet highly effective control architecture which easily scales for systems with a large number of evaporators. The ability of the proposed control architecture and design to provide both high performance and reduced energy consumption is demonstrated through a simulated case study.

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1. Introduction

In an effort to reduce energy use in buildings, there is a growing interest in the performance and efficiency of large-scale heating and cooling systems. In particular, variable-refrigerant-flow (VRF) systems, also termed multi-evaporator vapor-compression systems, are gaining significant market attention in the U.S. because of the benefits they offer with respect to energy efficiency and flexibility for architects [1,2]. VRF systems consist of a single condenser (heat rejection unit) and compressor which are then connected to n evaporators (heat absorbing units) (Fig. 1).

For practical acceptance, both in design and installation, it is critical to have a control architecture that is scalable ($n \gg 2$) as well as accessible to the building HVAC installation practitioner. It is common to have more than 30 evaporators interacting with the single condenser and compressor in a VRF system [1]. Each zone in a building typically has a dedicated evaporator, electronic expansion valve (EEV), and evaporator fan which are used to deliver a desired amount of cooling to that zone, so a decentralized control architecture is more practical for scalability as the number of evaporators increases. However, because all n evaporators are interacting with a single compressor and condenser, a decentralized approach must be robust to the coupling which exists between evaporators in the overall network. In this paper we present and validate a partially decentralized control architecture which is scalable to arbitrarily large VRF systems.

The first challenge in controlling VRF systems is deriving an appropriate model of the system dynamics. Black-box models obtained from traditional data-driven system identification techniques suffer from various disadvantages: (1) they cannot be developed during the system design phase since the use of black-box model identification techniques requires that the system already exists and (2) they are not scalable (with respect to the number of evaporators in the system) since an identified model will be specific to the system configuration at the time data is gathered. *Therefore, our first objective is to develop a generalized model for an n -evaporator VRF system which overcomes these disadvantages of black-box modeling approaches but is still suitable for model-based control design.* We will do this using a gray-box modeling approach and by exploiting the one-way coupling between the VRF system fluid and thermal dynamics to derive separate models for each class of dynamics.

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Nomenclature

Symbols

a	aperture
C	capacitance
$\dot{m}$	mass flow rate
P	pressure
$\dot{Q}$	heat transfer rate
R	resistance
T	temperature
$K, \alpha, \beta, \eta, \lambda, \mu$	coefficient of linearization
ω	rotational speed

Subscripts

a	air
c	condenser
e	evaporators
f	fan
F	fluid dynamics
k	compressor
L	load
q	junction
r	refrigerant
T	thermal dynamics
v	valve
w	coil wall

The second challenge is to design an appropriate control architecture for large-scale VRF systems. Various control schemes have been proposed in the literature for dual- and triple-evaporator VRF systems. In [4] a cascaded control structure is proposed for a dual-evaporator VRF system which takes advantage of the time scale separation between the refrigerant thermal dynamics and the zone air thermal dynamics. The paper [5] introduces a fuzzy algorithm used with PID controllers to regulate zone air temperatures for a triple-evaporator VRF system. However, neither of these approaches is shown to be scalable to systems larger than $n=2$ or $n=3$, respectively. In [6] a dual-evaporator VRF system with discharge pressure regulating (DPR) valves at the outlet of each evaporator is analyzed and then controlled using a decentralized approach which achieves a desired cooling capacity and evaporator superheat in each evaporator. The presence of DPR valves decouples evaporator pressures in the system thereby significantly reducing the dynamic coupling and enabling a decentralized control architecture. However, while effective for a dual-evaporator VRF system, [6] does not offer additional analysis to understand how their approach might scale to larger VRF systems. What is clearly lacking in the literature is a scalable dynamical analysis of *arbitrarily large* VRF systems, particularly those *without* DPR valves (which are often omitted due to the variable cost they add from the manufacturer's perspective). *Therefore, our second objective is to propose and implement a scalable control architecture for VRF systems which is guided by an analysis of the VRF system dynamics and is amenable to industrial implementation.*

This paper is organized as follows. In Section 2, the dynamic models for the fluid dynamics and thermal dynamics are derived using electrical circuit analogies. In Section 3, a partially decentralized control architecture is proposed which uses separate controllers to meet

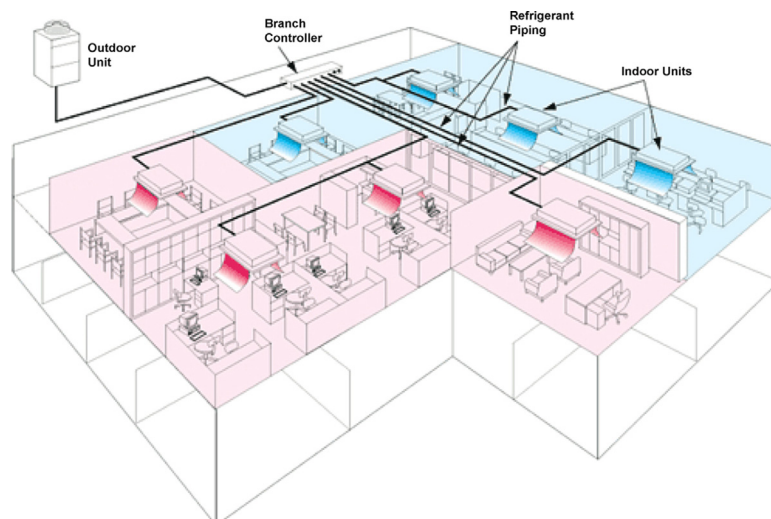


Fig. 1. Schematic of a Mitsubishi Electric VRF system [3].

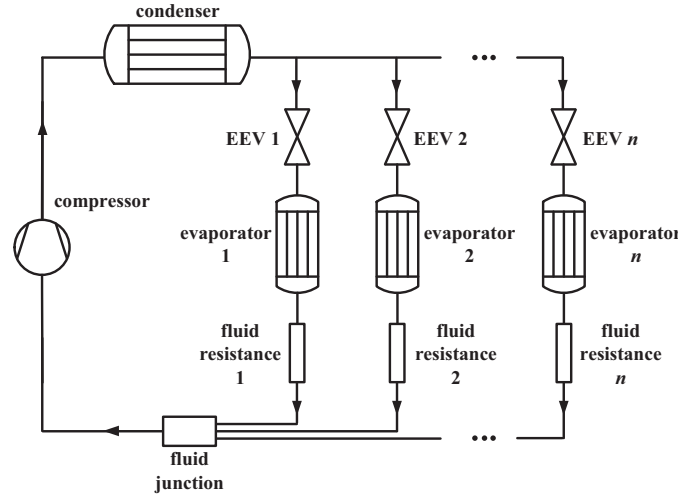


Fig. 2. Schematic of VRF system for fluid dynamics analysis.

the global and local performance objectives of the VRF system. Local and global controller designs are subsequently presented in Section 4. A case study is presented in Section 5, and conclusions of this work are summarized in Section 6.

2. Dynamic modeling

In this section, electrical circuit analogies are used to derive linearized models of the fluid dynamics and thermal dynamics of a standard VRF system. Although these dynamics are highly nonlinear, linearized models are typically used for the purpose of control design. An advantage of this gray-box linearized modeling approach is that the functional relationships between the physical parameters are conserved, and therefore, the models presented are valid around different operating conditions (albeit with different parameter values) [7,8]. Two linearized models are derived – one each for the fluid dynamics and the thermal dynamics in the VRF system. Therefore, all variables in the subsequent derivations represent deviation variables with respect to some nominal operating condition.

A schematic depicting a standard VRF system architecture is shown in Fig. 2. The basic operation of a standard VRF system is as follows. First, superheated refrigerant vapor is compressed to a higher temperature and pressure. The still superheated refrigerant vapor then flows through the condenser where it rejects thermal energy to the ambient air passing over the condenser coil. The refrigerant leaves the condenser as a subcooled liquid and is subsequently divided through n different flow paths each leading to one EEV and evaporator pair. The i th refrigerant stream undergoes a rapid drop in pressure and temperature through the i th EEV and then enters the i th evaporator as a two-phase fluid where it absorbs thermal energy from the zone air passing over the evaporator coil. The n refrigerant streams recombine downstream of the n evaporators, and the single refrigerant stream then enters the compressor to repeat the cycle.

2.1. Fluid dynamics

The electrical circuit analogy for the fluid dynamics is shown in Fig. 3. In this modeling approach, refrigerant mass flow rate is analogous to electric current and pressure differential is analogous to voltage potential. It follows, then, that the condenser, evaporators, and the fluid junction are modeled as capacitors because they store refrigerant. Moreover, the EEVs and fluid resistances are modeled as resistors because they are dissipative devices with a pressure drop across them. These component-level analogs are summarized in Table 1.

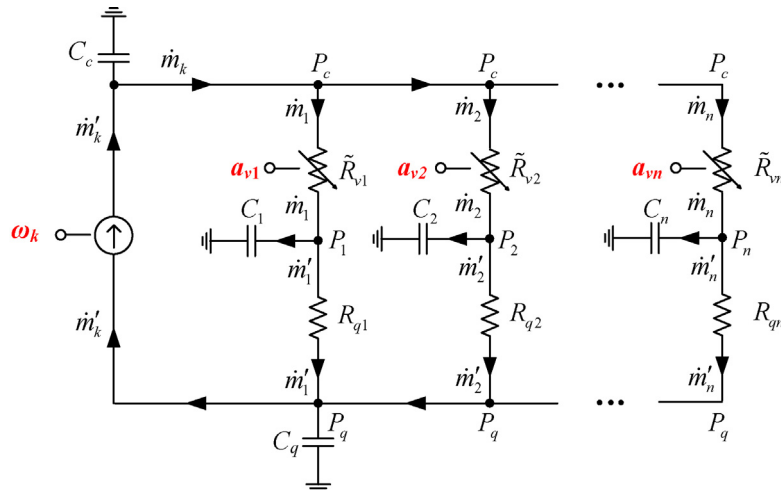


Fig. 3. Electrical circuit analogy of the fluid dynamics of a standard VRF system. The symbol $\bigcirc$ — indicates the presence of a control input.

Table 1
Electrical circuit analogs for VRF systems.

VRF system	Electrical circuit analog
Refrigerant mass flow rate	Electric current
Pressure differential	Voltage potential
Fluid resistance	Resistor
Compressor	Current source
EEV	Variable resistor
Condenser/evaporators	Capacitor
Fluid junction	Capacitor

There are n evaporators in the system, denoted by the set $N = \{1, \dots, n\}$. There are $n+2$ differential equations which correspond to the $n+1$ heat exchangers in the system (one condenser and n evaporators) and the flow junction downstream of the n evaporators:

$$C_c \dot{P}_c = \dot{m}'_k - \dot{m}_k, \quad (1)$$

$$C_i \dot{P}_i = \dot{m}_i - \dot{m}'_i \quad \forall i \in N, \quad (2)$$

$$C_q \dot{P}_q = \sum_{i=1}^n \dot{m}'_i - \dot{m}'_k. \quad (3)$$

The dynamic states of the VRF system are the refrigerant pressures, P_c, P_i for $i \in N$ and P_q . In (1)–(3), $\dot{P}$ is the derivative of the state, not to be confused with $\dot{m}$, refrigerant mass flow rate, which is an algebraic quantity in this system.

As a mass flow device, the compressor is modeled as a current source which provides a refrigerant mass flow rate, $\dot{m}'_k$, to the system. The mass flow rate, given by

$$\dot{m}'_k = \beta_{k1} P_q + \beta_{k2} P_c + \beta_{k3} \omega_k, \quad (4)$$

is modeled as a linear function of the pressures at the inlet and outlet of the compressor as well as the compressor speed, ω_k (a control input to the system). The parameters β_{k1} , β_{k2} , and β_{k3} , are linearization coefficients.

Each EEV is modeled as a variable resistor whose pressure drop (analogous to voltage potential) is a function of a resistance $\tilde{R}_{vi}$ which in turn is a nonlinear function of the EEV aperture, a_{vi} (a control input to the system). Thus, from Fig. 3 we obtain

$$P_c - P_i = \tilde{R}_{vi}(a_{vi}) \dot{m}_i \quad \forall i \in N. \quad (5)$$

For simplicity, (5) is linearized so that the total pressure drop across each EEV is expressed as a linear combination of the refrigerant mass flow rate through the EEV and the EEV aperture, which yields

$$P_c - P_i = R_{vi} \dot{m}_i - K_{vi} a_{vi} \quad \forall i \in N. \quad (6)$$

Both R_{vi} and K_{vi} are positive constants arising from the linearization; R_{vi} can be thought of as the nominal resistance associated with the EEV. Following Kirchhoff's current law, the n refrigerant streams exiting the evaporators will combine through a junction near the inlet to the compressor. As shown in Fig. 3, the combined refrigerant stream is at a common pressure, P_q , where $P_q \leq P_i \quad \forall i \in N$. We model the pressure drop downstream of each evaporator using a resistor with resistance R_{qi} , $i \in N$, to obtain

$$P_i - P_q = R_{qi} \dot{m}'_i \quad \forall i \in N. \quad (7)$$

Finally, we can write one mass flow conservation equation for the VRF system which is given by

$$\dot{m}_k = \dot{m}_1 + \dot{m}_2 + \dots + \dot{m}_{n-1} + \dot{m}_n. \quad (8)$$

The governing equations have been written so that all of the system capacitances, resistances, and linearized coefficients are positive. The resulting state-space system representation $\mathbf{S}_F : \delta \dot{\mathbf{x}}_F = \mathbf{A}_F \delta \mathbf{x}_F + \mathbf{B}_F \delta \mathbf{u}_F$, $\mathbf{A}_F \in \mathbb{R}^{(n+2) \times (n+2)}$, $\mathbf{B}_F \in \mathbb{R}^{(n+2) \times (n+1)}$, is shown in (72)¹ where

$$\begin{aligned} \delta \mathbf{x}_F &= [P_1 \ P_2 \ \dots \ P_n \ P_c \ P_q]^T \in \mathbb{R}^{n+2}, \\ \delta \mathbf{u}_F &= [a_{v1} \ a_{v2} \ \dots \ a_{vn} \ \omega_k]^T \in \mathbb{R}^{n+1}. \end{aligned} \quad (9)$$

Now we provide an important result regarding the controllability of (72).

Proposition 1. *The linear state-space model for the VRF system fluid dynamics given in (72) is not controllable.*

Proof. One can verify that the vector

$$\nu = \begin{bmatrix} \frac{C_1}{C_q} & \frac{C_2}{C_q} & \dots & \frac{C_n}{C_q} & \frac{C_c}{C_q} & 1 \end{bmatrix}$$

is in the left nullspace of the matrix $[\mathbf{A}_F - \lambda \mathbf{I} | \mathbf{B}_F]$ for $\lambda = 0$. Therefore the system has an uncontrollable eigenvalue at zero. $\square$

Remark 1. The physical interpretation for this result can be obtained upon closer examination of (1)–(3) and (8). By virtue of the mass flow conservation equation in (8), the sum of the $n+2$ differential equations in (1)–(3) is $C_q \dot{P}_q = - (C_c \dot{P}_c + \sum_{i=1}^n C_i \dot{P}_i)$. It follows that

¹ The state-space model representation is located at the end of the paper because of the size of the $\mathbf{A}_F$ and $\mathbf{B}_F$ matrices.

$C_q P_q = -\left(C_c P_c + \sum_{i=1}^n C_i P_i\right) + \gamma$ where γ is a constant of integration that depends on the initial state of the system, assumed here to equal 0.

Since the junction pressure, P_q , can be expressed as a linear combination of the n evaporator pressures and the condenser pressure, we eliminate P_q from the system model. With an abuse of notation, we will denote the revised state-space representation as $\mathbf{S}_F : \delta \dot{\mathbf{x}}_F = \mathbf{A}_F \delta \mathbf{x}_F + \mathbf{B}_F \delta \mathbf{u}_F$ where $\mathbf{A}_F, \mathbf{B}_F \in \mathbb{R}^{(n+1) \times (n+1)}$ are shown in (73), and

$$\begin{aligned} \delta \mathbf{x}_F &= \begin{bmatrix} P_1 & P_2 & \cdots & P_n & P_c \end{bmatrix}^T \in \mathbb{R}^{n+1}, \\ \delta \mathbf{u}_F &= \begin{bmatrix} a_{v1} & a_{v2} & \cdots & a_{vn} & \omega_k \end{bmatrix}^T \in \mathbb{R}^{n+1}. \end{aligned} \quad (10)$$

As will be discussed in more detail in Section 3, we are primarily interested in a decentralized control architecture to meet the local performance objective of the i th evaporator providing cooling to the i th building zone. Therefore, let us consider the dynamics associated with the n refrigerant pressures, P_i for $i \in N$ and provide a useful result regarding the dynamic coupling between them.

Proposition 2. The top left $n \times n$ block of $\mathbf{A}_F$ is diagonally dominant if

$$C_q \geq \left(1 + \frac{R_{qi}}{R_{vi}}\right)^{-1} \sum_{\substack{j=1 \\ j \neq i}}^n C_j \quad \forall i \in N. \quad (11)$$

Proof. The $n \times n$ block of matrix $\mathbf{A}_F$ is diagonally dominant if $|a_{ii}| \geq \sum_{j \neq i} |a_{ij}|$ for $i, j \in N$ where a_{ij} are the elements of $\mathbf{A}_F$. Substituting for a_{ii} and a_{ij} using the $n \times n$ block of (73) gives

$$\left| -\frac{1}{C_i} \left(\frac{1}{R_{vi}} + \frac{1}{R_{qi}} \right) \right| \geq \sum_{\substack{j=1 \\ j \neq i}}^n \left| -\frac{C_j}{C_q C_i R_{qi}} \right|, \quad (12)$$

from which the result (11) follows. $\square$

For building VRF systems, it is typical that $C_i \ll C_q \forall i \in N$ because C_i represents the fluid capacitance of the refrigerant in a single evaporator whereas C_q represents the fluid capacitance of the n refrigerant streams combined in the junction near the inlet of the compressor. Therefore (11) is easily satisfied for VRF systems as will be shown in Table 4. Moreover, from (73) it is evident that the i th EEV input, a_{vi} , only directly affects the i th evaporator pressure, P_i . Motivated by these observations, we will define a subsystem representation for the fluid dynamics of the i th evaporator pressure (when considered in isolation) as

$$\mathbf{S}_{Fi} : \dot{P}_i = -\frac{1}{C_i} \left(\frac{1}{R_{vi}} + \frac{1}{R_{qi}} \right) P_i + \frac{K_{vi}}{C_i R_{vi}} a_{vi}. \quad (13)$$

The overall dynamics for the i th evaporator pressure (including coupling with the other evaporators) are defined as

$$\mathbf{S}_F : \dot{P}_i = -\frac{1}{C_i} \left(\frac{1}{R_{vi}} + \frac{1}{R_{qi}} \right) P_i + \frac{K_{vi}}{C_i R_{vi}} a_{vi} + \sum_{\substack{j=1 \\ j \neq i}}^n \frac{-C_j}{C_q C_i R_{qi}} P_j + \frac{1}{C_i} \left(\frac{1}{R_{vi}} - \frac{C_c}{C_q R_{qi}} \right) P_c. \quad (14)$$

Remark 2. The dynamics of P_c evolve more slowly than those of P_i for $i \in N$ because C_i represents the refrigerant fluid capacitance in a single evaporator whereas C_c represents the fluid capacitance of the n refrigerant streams combined in the condenser. Therefore, the effect of P_c on each of the evaporator pressures, P_i , will be treated as a slowly varying disturbance. This assumption will be verified later on in Section 5.1.1.

2.2. Thermal dynamics

A schematic depicting a standard VRF system for analysis of its thermal dynamics is shown in Fig. 4. The electrical circuit analogy for the thermal dynamics is shown in Fig. 5. In this modeling approach, heat transfer rate is analogous to electric current and temperature differential is analogous to voltage potential. It follows, then, that the condenser and evaporator coil walls and the thermal junction are modeled as capacitors because they store heat. Moreover, the EEVs and thermal resistances are modeled as resistors because they are dissipative devices with a temperature drop across them. These component-level analogs are summarized in Table 2.

Table 2
Electrical circuit analogs for thermal dynamics of VRF systems.

VRF system	Electrical circuit analog
Heat transfer rate	Electric current
Temperature differential	Voltage potential
Thermal resistance	Resistor
Compressor	Voltage source and current source
EEV	Variable resistor
Condenser/evaporator coil wall	Thermal capacitor
Thermal junction	Thermal capacitor

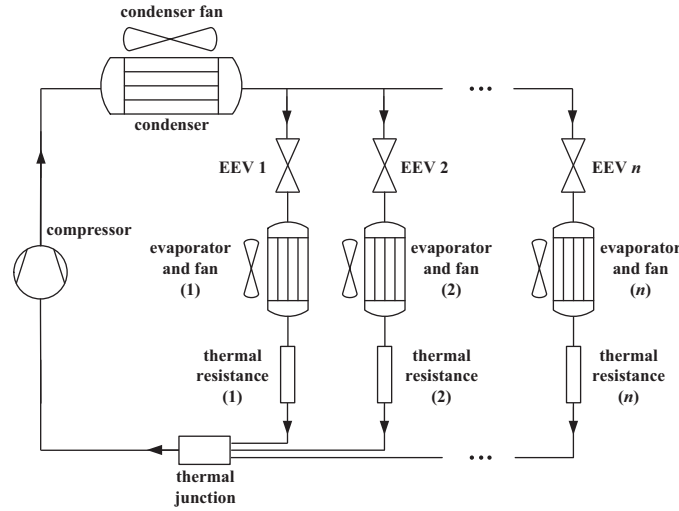


Fig. 4. Schematic of VRF system for thermal dynamics analysis.

Heat exchangers act as thermal capacitors because of their large metal coil which stores heat during system operation. The first $n + 1$ differential equations correspond to the thermal capacitance of the $n + 1$ heat exchangers in the VRF system (one condenser and n evaporators). A lumped coil wall temperature state is defined for each heat exchanger as described by

$$\begin{aligned} C_{wc} \dot{T}_{wc} &= \dot{Q}_{rwc} - \dot{Q}_{wac}, \\ C_{wi} \dot{T}_{wi} &= \dot{Q}_{awi} - \dot{Q}_{wri} \quad \forall i \in N. \end{aligned} \quad (15)$$

In (15), $\dot{T}$ is the derivative of the state, not to be confused with $\dot{Q}$, heat transfer rate, which is an algebraic quantity in this system. The refrigerant streams exiting each evaporator meet at a junction downstream of the n evaporators. The temperature of the combined refrigerant stream, T_{rq} , is defined as an additional state in the system with dynamics given by

$$C_{rq} \dot{T}_{rq} = \sum_i^n \dot{Q}'_{ri} - \dot{Q}_{re} + \kappa_{rq} \sum_i^n \dot{m}'_i - \kappa_{rq} \dot{m}'_k. \quad (16)$$

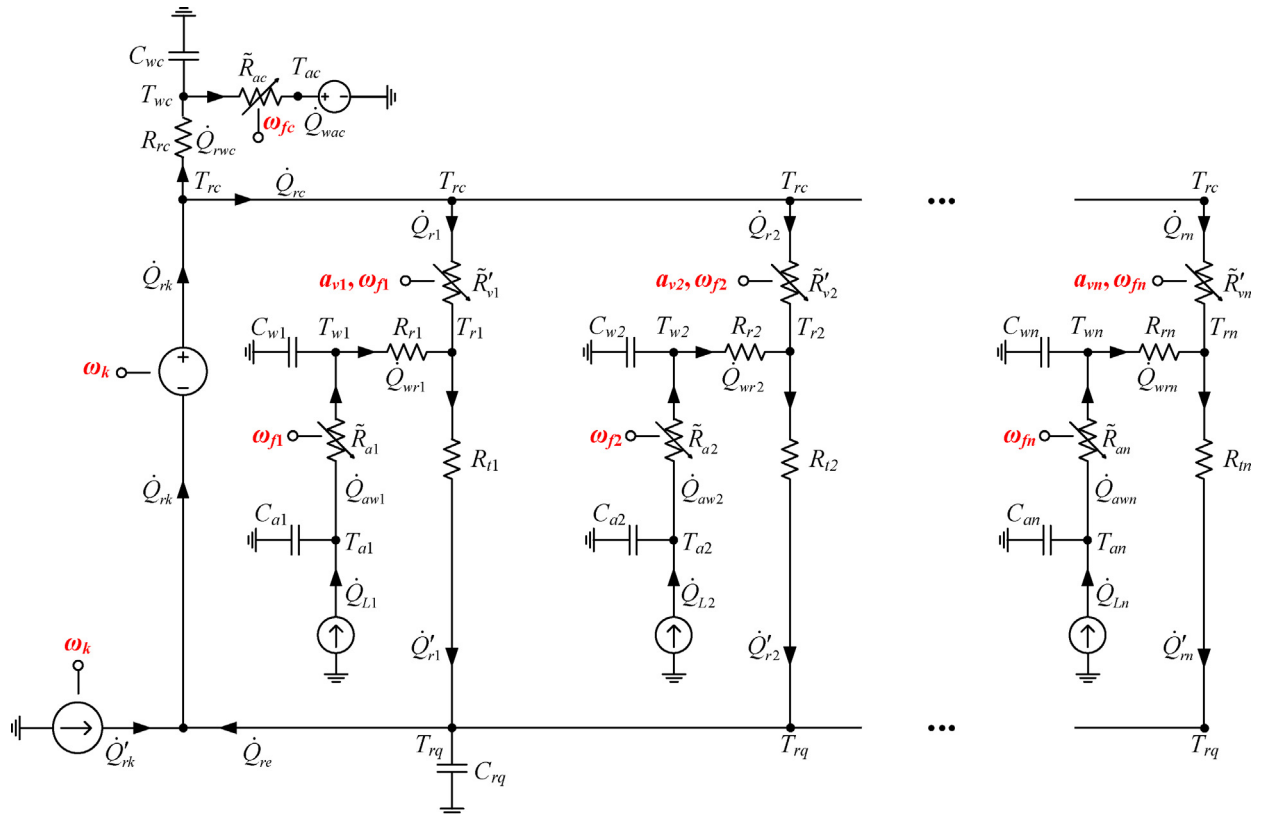


Fig. 5. Electrical circuit analogy of the thermal dynamics of a standard VRF system. The symbol $\bigcirc$ — indicates the presence of a control input.

The constant C_{rq} represents the thermal capacitance of the refrigerant inside the junction where the individual streams mix together, and κ_{rq} is a linearization coefficient. The remaining n dynamic states are defined as the air temperatures inside each of the n zones of the building, T_{ai} , $i \in N$, with dynamics given by

$$C_{ai}\dot{T}_{ai} = \dot{Q}_{Li} - \dot{Q}_{awi} \quad \forall i \in N, \quad (17)$$

where $\dot{Q}_{Li}$, $i \in N$, are unknown load disturbances to each zone. Typically, VRF control designers do not have knowledge of the building dynamics which will ultimately interact with the VRF system. Therefore we assume that there is no heat transfer between adjacent zones of the building.

Associated with each heat exchanger are an air-side thermal resistance and a refrigerant-side thermal resistance. The air-side thermal resistance for each heat exchanger varies nonlinearly with heat exchanger fan speed:

$$\begin{aligned} T_{wc} - T_{ac} &= \dot{Q}_{wac} \tilde{R}_{fc}(\omega_{fc}), \\ T_{ai} - T_{wi} &= \dot{Q}_{awi} \tilde{R}_{fi}(\omega_{fi}) \quad \forall i \in N. \end{aligned} \quad (18)$$

For simplicity, the equations in (18) are linearized as

$$\begin{aligned} T_{wc} - T_{ac} &= R_{ac} \dot{Q}_{wac} - K_{fc} \omega_{fc}, \\ T_{ai} - T_{wi} &= R_{ai} \dot{Q}_{awi} - K_{fi} \omega_{fi} \quad \forall i \in N, \end{aligned} \quad (19)$$

so that the heat transfer rate between the air and wall in each heat exchanger is approximated as a linear function of the temperature difference between the air and the wall, and the corresponding fan speed (a control input to the system). Both R_{ai} and K_{fi} are positive constants arising from the linearization; R_{ai} can be thought of as the nominal thermal resistance between the air and wall. The previous statement also holds for R_{ac} and K_{fc} .

The refrigerant-side thermal resistance between the refrigerant and the coil wall in the condenser ($\tilde{R}_{rc}$) and each evaporator ($\tilde{R}_{ri}$, $\forall i \in N$) varies nonlinearly with refrigerant mass flow rate:

$$\begin{aligned} T_{rc} - T_{wc} &= \dot{Q}_{rwc} \tilde{R}_{rc}(\dot{m}'_k), \\ T_{wi} - T_{ri} &= \dot{Q}_{wri} \tilde{R}_{ri}(\dot{m}_i) \quad \forall i \in N. \end{aligned} \quad (20)$$

Again the equations in (20) are linearized as

$$\begin{aligned} T_{rc} - T_{wc} &= R_{rc} \dot{Q}_{rwc} - \lambda_{rc} \dot{m}'_k, \\ T_{wi} - T_{ri} &= R_{ri} \dot{Q}_{wri} - \lambda_{ri} \dot{m}_i \quad \forall i \in N, \end{aligned} \quad (21)$$

so that the heat transfer rate between the refrigerant and wall in each heat exchanger is approximated as a linear function of the temperature difference between the refrigerant and the wall as well as the refrigerant mass flow rate through the heat exchanger coil. Both R_{ri} and λ_{ri} are positive constants arising from the linearization; R_{ri} can be thought of as the nominal thermal resistance between the refrigerant and coil wall in the i th evaporator. An analogous statement can be made for R_{rc} and λ_{rc} with regards to the condenser.

The temperatures T_{ri} , $i \in N$, and T_{rc} are approximated as

$$\begin{aligned} T_{rc} &= \eta_{rc} P_c, \\ T_{ri} &= \eta_{ri} P_i \quad \forall i \in N, \end{aligned} \quad (22)$$

where η_{rc} and η_{ri} are linearization coefficients.

The compressor is modeled as both a current source and a voltage source which provide (respectively) additional thermal energy to, and a temperature differential across, the refrigerant flowing through the compressor. The additional heat transfer rate is given by

$$\dot{Q}'_{rk} = \alpha_{k1} T_{rc} - \alpha_{k2} T_{rq} + \alpha_{k3} \omega_k, \quad (23)$$

and is a function of the temperatures at the inlet and outlet of the compressor as well as the compressor speed, ω_k . The parameters α_{k1} , α_{k2} , and α_{k3} , are linearization coefficients. The net heat transfer rate of the refrigerant exiting the compressor is modeled as

$$\dot{Q}_{rk} = \dot{Q}_{re} + \dot{Q}'_{rk} = \gamma_{k1} T_{rc} + \gamma_{k2} T_{rq} + \gamma_{k3} \omega_k, \quad (24)$$

where the parameters γ_{k1} , γ_{k2} , and γ_{k3} are linearization coefficients.

Each EEV is modeled as a variable resistor whose thermal resistance, $\tilde{R}'_{vi}$, is a function of the corresponding EEV aperture, a_{vi} , and the evaporator fan speed, ω_{fi} . Thus, from Fig. 5 we obtain

$$T_{rc} - T_{ri} = \tilde{R}'_{vi}(a_{vi}, \omega_{fi}) \dot{Q}_{ri} \quad \forall i \in N, \quad (25)$$

which is linearized so that the rate at which heat is being carried by the refrigerant moving through the EEV is expressed as a linear combination of the temperature drop across the EEV, the EEV aperture, and the evaporator fan speed. This produces

$$T_{rc} - T_{ri} = R'_{vi} \dot{Q}_{ri} - K'_{vi} a_{vi} - K'_{ki} \omega_{fi} \quad \forall i \in N, \quad (26)$$

where the coefficients R'_{vi} , K'_{vi} , and K'_{ki} are positive constants arising from the linearization; R'_{vi} can be thought of as the nominal thermal resistance associated with the EEV.

The temperature change downstream of each evaporator is modeled using the following equation:

$$T_{ri} - T_{rq} = \mu_{m1i} \dot{Q}'_{ri} - \mu_{m2i} \dot{m}'_i - \mu_{m3i} \dot{Q}_{re} + \mu_{m4i} \dot{m}'_k \quad \forall i \in N. \quad (27)$$

Since the sign of the temperature change between T_{ri} and T_{rq} is not fixed, we do not explicitly model the temperature change as a resistor or as a voltage source. Finally, we can write two energy conservation equations for the VRF system modeled in Fig. 5 which are given by

$$\begin{aligned}\dot{Q}_c &= \dot{Q}_{r1} + \dot{Q}_{r2} + \dots + \dot{Q}_{rn}, \\ \dot{Q}_{rk} &= \dot{Q}'_{rk} + \dot{Q}_{re}.\end{aligned}\quad (28)$$

The governing equations have been written so that all of the system capacitances, resistances, and linearized coefficients are positive. The resulting state-space system representation is given by

$$\mathbf{S}_T : \delta \dot{\mathbf{x}}_T = \mathbf{A}_T \delta \mathbf{x}_T + [\mathbf{B}_{T1} \quad \mathbf{B}_{T2}] \begin{bmatrix} \delta \mathbf{u}_F \\ \delta \mathbf{u}_{T2} \end{bmatrix} + [\mathbf{W}_{T1} \quad \mathbf{W}_{T2}] \begin{bmatrix} \delta \mathbf{x}_F \\ \delta \mathbf{d}_{T2} \end{bmatrix}, \quad (29)$$

where

$$\begin{aligned}\delta \mathbf{x}_T &= [T_{w1} \quad T_{w2} \quad \dots \quad T_{wn} \quad T_{a1} \quad T_{a2} \quad \dots \quad T_{an} \quad T_{wc} \quad T_{rq}]^T \in \mathbb{R}^{2n+2}, \\ [\delta \mathbf{u}_F \quad \delta \mathbf{u}_T]^T &= [a_{v1} \quad a_{v2} \quad \dots \quad a_{vn} \quad \omega_k \quad \omega_{f1} \quad \omega_{f2} \quad \dots \quad \omega_{fn} \quad \omega_{fc}]^T \in \mathbb{R}^{2n+2}, \\ [\delta \mathbf{x}_F \quad \delta \mathbf{d}_{T2}]^T &= [P_1 \quad P_2 \quad \dots \quad P_n \quad P_c \quad \dot{Q}_{L1} \quad \dot{Q}_{L2} \quad \dots \quad \dot{Q}_{Ln} \quad T_{ac}]^T \in \mathbb{R}^{2n+2}.\end{aligned}\quad (30)$$

Recall that the states and inputs are deviation variables with respect to the point about which the system is linearized.

From a controls perspective, we are primarily interested in regulating each zone air temperature, T_{ai} , which is also dependent on the evaporator coil wall temperature, T_{wi} . It can be verified from (15) and (16) that the state derivatives $\dot{T}_{wc}$ and $\dot{T}_{rq}$ depend only on their own states, T_{wc} and T_{rq} , respectively. Therefore, we can consider a reduced state space representation for the thermal dynamics which excludes the states T_{wc} and T_{rq} . With an abuse of notation, we will denote the revised state-space representation as

$$\mathbf{S}_T : \delta \dot{\mathbf{x}}_T = \mathbf{A}_T \delta \mathbf{x}_T + \mathbf{B}_T \delta \mathbf{u}_T + [\mathbf{W}_{T1} \quad \mathbf{W}_{T2}] \begin{bmatrix} \delta \mathbf{d}_{T1} \\ \delta \mathbf{d}_{T2} \end{bmatrix}, \quad (31)$$

where

$$\begin{aligned}\delta \mathbf{x}_T &= [T_{a1} \quad T_{w1} \quad T_{a2} \quad T_{w2} \quad \dots \quad T_{an} \quad T_{wn}]^T \in \mathbb{R}^{2n}, \\ \delta \mathbf{u}_T &= [\omega_{f1} \quad a_{v1} \quad \omega_{f2} \quad a_{v2} \quad \dots \quad \omega_{fn} \quad a_{vn}]^T \in \mathbb{R}^{2n}, \\ [\delta \mathbf{d}_{T1} \quad \delta \mathbf{d}_{T2}]^T &= [\dot{Q}_{L1} \quad P_1 \quad \dot{Q}_{L2} \quad P_2 \quad \dots \quad \dot{Q}_{Ln} \quad P_n \quad P_c]^T \in \mathbb{R}^{2n+1}.\end{aligned}\quad (32)$$

Note that the state, input, and disturbance vectors have been rearranged to highlight the natural decoupling between the evaporators. The matrices $\mathbf{A}_T$, $\mathbf{B}_T$, $\mathbf{W}_{T1}$, and $\mathbf{W}_{T2}$ are given in (74).

Remark 3. From (74) it can be seen that both $\mathbf{A}_T$ and $\mathbf{B}_T$ are block diagonal. This observation motivates the use of a decentralized control approach for the control of each zone air temperature, T_{ai} , using the following subsystem representation for the thermal dynamics associated with the i th evaporator and building zone:

$$\mathbf{S}_{Ti} : \begin{bmatrix} \dot{T}_{ai} \\ \dot{T}_{wi} \end{bmatrix} = \begin{bmatrix} -\frac{1}{C_{ai}R_{ai}} & \frac{1}{C_{ai}R_{ai}} \\ \frac{1}{C_{wi}R_{ai}} & -\frac{1}{C_{wi}} \left(\frac{1}{R_{ai}} + \frac{1}{R_{ri}} \right) \end{bmatrix} \begin{bmatrix} T_{ai} \\ T_{wi} \end{bmatrix} + \begin{bmatrix} -\frac{K_{fi}}{C_{ai}R_{ai}} & 0 \\ \frac{K_{fi}}{C_{wi}R_{ai}} & -\frac{K_{vi}\lambda_{ri}}{C_{wi}R_{ri}R_{vi}} \end{bmatrix} \begin{bmatrix} \omega_{fi} \\ a_{vi} \end{bmatrix} + \begin{bmatrix} \frac{1}{C_{ai}} & 0 \\ 0 & \frac{\eta_{r1i}}{C_{wi}R_{ri}} + \frac{\lambda_{ri}}{C_{wi}R_{ri}R_{vi}} \end{bmatrix} \begin{bmatrix} \dot{Q}_{Li} \\ P_i \end{bmatrix} \quad \forall i \in N. \quad (33)$$

2.3. Complete system model

As mentioned earlier, there exists a one-way coupling between the fluid and thermal dynamics. Based on Proposition 2 and Remark 3, we combine (13) and (33) and define the subsystem representation for the dynamics of the i th evaporator and corresponding building zone as

$$\mathbf{S} : \dot{\mathbf{x}}_i = \mathbf{A}_i \mathbf{x}_i + \mathbf{B}_i \mathbf{u}_i + \sum_{\substack{j=1 \\ j \neq i}}^n \mathbf{A}_{ij} \mathbf{x}_j + \mathbf{W}_i \mathbf{d}_i \quad \forall i \in N, \quad (34)$$

where $\sum_{j=1}^n \mathbf{A}_{ij} \mathbf{x}_j$ represents the coupling between the n evaporators. With the thermal subsystems in (33) being completely decoupled, the only source of coupling comes from the fluid subsystems in (13). Thus,

$$\mathbf{A}_{ij} = \begin{bmatrix} 0_{2 \times 2} & 0_{2 \times 1} \\ 0_{1 \times 2} & -\frac{C_j}{C_q C_i R_{qi}} \end{bmatrix}. \quad (35)$$

For decentralized control, a decoupled subsystem representation defined as

$$\mathbf{S}_i : \dot{\mathbf{x}}_i = \mathbf{A}_i \mathbf{x}_i + \mathbf{B}_i \mathbf{u}_i + \mathbf{W}_i \mathbf{d}_i \quad \forall i \in N, \quad (36)$$

will be used for control design of the local controllers to be presented in the next section. The subsystem in (36) is expanded as follows (where $\mathbf{x}_i \in \mathbb{R}^3$, $\mathbf{u}_i \in \mathbb{R}^2$, and $\mathbf{d}_i \in \mathbb{R}^2$):

$$\mathbf{S}_i : \begin{bmatrix} \dot{T}_{ai} \\ \dot{T}_{wi} \\ \dot{P}_i \end{bmatrix} = \begin{bmatrix} -\frac{1}{C_{ai}R_{ai}} & \frac{1}{C_{ai}R_{ai}} & 0 \\ \frac{1}{C_{wi}R_{ai}} & -\frac{1}{C_{wi}} \left(\frac{1}{R_{ai}} + \frac{1}{R_{ri}} \right) & \frac{\eta_{r1i}}{C_{wi}R_{ri}} + \frac{\lambda_{ri}}{C_{wi}R_{ri}R_{vi}} \\ 0 & 0 & -\frac{1}{C_i} \left(\frac{1}{R_{vi}} + \frac{1}{R_{qi}} \right) \end{bmatrix} \begin{bmatrix} T_{ai} \\ T_{wi} \\ P_i \end{bmatrix} + \begin{bmatrix} -\frac{K_{fi}}{C_{ai}R_{ai}} & 0 \\ \frac{K_{fi}}{C_{wi}R_{ai}} & -\frac{K_{vi}\lambda_{ri}}{C_{wi}R_{ri}R_{vi}} \\ 0 & \frac{K_{vi}}{C_i R_{vi}} \end{bmatrix} \begin{bmatrix} \omega_{fi} \\ a_{vi} \end{bmatrix} + \begin{bmatrix} \frac{1}{C_{ai}} & 0 \\ 0 & 0 \\ 0 & \frac{1}{C_i} \left(\frac{1}{R_{vi}} - \frac{C_c}{C_q R_{qi}} \right) \end{bmatrix} \begin{bmatrix} \dot{Q}_{Li} \\ P_c \end{bmatrix} \quad \forall i \in N \quad (37)$$

The n subsystems can be written in compact form for the coupled system $\mathbf{S}$ and the decoupled system $\mathbf{S}_D$ as

$$\mathbf{S} : \dot{\mathbf{x}} = \mathbf{A}_D \mathbf{x} + \mathbf{B}_D \mathbf{u} + \mathbf{A}_C \mathbf{x} + \mathbf{W}_D \mathbf{d}, \quad (38)$$

$$\mathbf{S}_D : \dot{\mathbf{x}} = \mathbf{A}_D \mathbf{x} + \mathbf{B}_D \mathbf{u} + \mathbf{W}_D \mathbf{d}. \quad (39)$$

3. Control architecture selection

To effectively control large-scale VRF systems, it is important to choose an appropriate control architecture that will address the control objectives of these systems. In particular, these objectives can be characterized as either being *local* or *global* in nature. Here we propose a partially decentralized control architecture which is comprised of (1) a set of n decentralized controllers designed to meet the *local* control objectives of each of the n evaporators and (2) a single controller designed to meet the *global* objectives of the VRF system.

As seen in Fig. 6, the global controller relies on measurements of both local states, $\mathbf{x}_L$, and global states, $\mathbf{x}_G$, but the n local controllers, $\mathbf{K}_i$ for $i \in N$, are completely decentralized from one another as well as from the global controller, $\mathbf{K}_G$. Based upon the dynamical analysis of the VRF system in the previous section, we define the following *local* state and input vectors as

$$\begin{aligned} \mathbf{x}_L &= [T_{a1} \ T_{w1} \ P_1 \ T_{a2} \ T_{w2} \ P_2 \ \cdots \ T_{an} \ T_{wn} \ P_n]^T \in \mathbb{R}^{3n}, \\ \mathbf{u}_L &= [\omega_{f1} \ a_{v1} \ \omega_{f2} \ a_{v2} \ \cdots \ \omega_{fn} \ a_{vn}]^T \in \mathbb{R}^{2n}, \end{aligned} \quad (40)$$

and the *global* state and input vectors as

$$\begin{aligned} \mathbf{x}_G &= [P_c \ T_{wc} \ T_{rq}]^T \in \mathbb{R}^3, \\ \mathbf{u}_G &= [\omega_k \ \omega_{fc}]^T \in \mathbb{R}^2. \end{aligned} \quad (41)$$

We define the n local performance objectives as regulating each of the building zone air temperatures to their respective desired values, $T_{ai,ref}$ for $i \in N$. This will be accomplished using the local control inputs, $\mathbf{u}_L$, in a decentralized manner.

At the global level, we define one control objective which addresses both performance and total energy consumption: the regulation of a *composite* superheat (which will be defined below). This will be accomplished using the global control inputs, $\mathbf{u}_G$. In *single*-evaporator

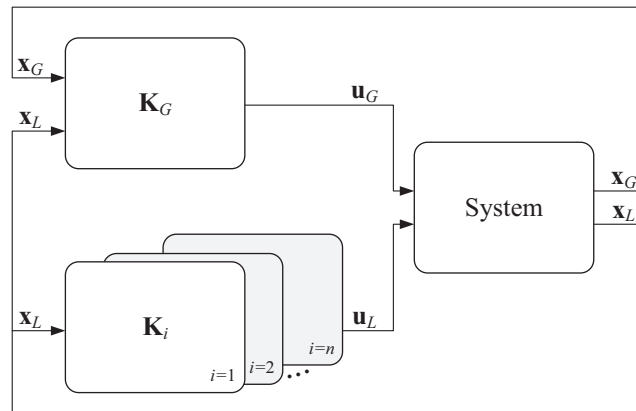


Fig. 6. Partially decentralized control architecture schematic.

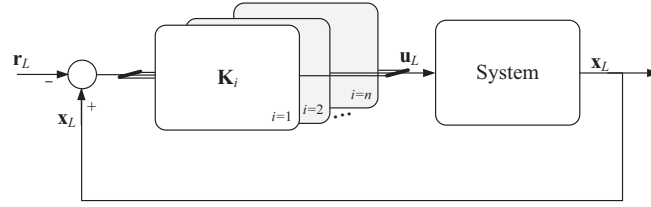


Fig. 7. Control architecture schematic for n local decentralized controllers.

vapor compression systems (VCSs), composite superheat is similar to evaporator superheat, defined as the temperature of the refrigerant (exiting the evaporator) above the saturation temperature. This is an important parameter in VCSs for two reasons. First, a positive value of evaporator superheat is desired to ensure that vapor (and no liquid) enters the compressor, thereby preventing compressor damage [9]. However, a very large value of evaporator superheat indicates that the evaporator is operating inefficiently because the heat transfer capabilities of the evaporator decrease significantly once the refrigerant is entirely evaporated. Therefore, to balance these performance and efficiency objectives, regulation of evaporator superheat to some small, nonnegative value is desirable [9].

In the context of VRF systems, we define the following *approximation* of the average of the n evaporator superheat values.

Definition 1. Composite superheat, T_{SH} , is defined as

$$T_{SH} = \frac{1}{n} \sum_{i=1}^n (T_{rq} - T_{ri}). \quad (42)$$

Through a simulated case study in Section 5, we will characterize the relationship between composite superheat and total energy consumption. We define total energy consumption as the combined energy consumption of the compressor, condenser fan, and n evaporator fans. We will also show that the regulation of composite superheat is necessary for ensuring that each evaporator meets its local performance objective in the presence of actuator saturation.

4. Controller design

A number of different control approaches can be used to design the n decentralized local controllers and the single global controller. As the control architecture from the previous section was designed in part due to its ease of implementation on a commercial VRF system, the individual controller design should also be accessible to industrial practitioners. Therefore, Linear-Quadratic-Regulator (LQR), an infinite horizon optimal control strategy, is used for the design of both the local and global controllers. For the decentralized local controllers, we will adopt an approach from [10] which applies LQR control design to a system of interconnected subsystems. It is worth noting that the global controller can be designed completely independently from the local controllers.

4.1. Local control design

The objective of each local controller is to regulate the desired zone air temperature, $T_{ai,ref}$, while rejecting a lumped load disturbance, $\dot{Q}_{Li}$. Motivated by the observations of Proposition 2 and Remark 3, a decentralized control architecture will be implemented in which the performance objective of the i th building zone will be achieved by the i th EEV and i th evaporator fan. The decoupled subsystem representation of the dynamics, S_i (36), will be used to design each local controller. This architecture is depicted schematically in Fig. 7.

To achieve regulation of T_{ai} with zero steady state error, each subsystem is augmented with an additional state defined as the integral of the error between T_{ai} and $T_{ai,ref}$. Moreover, since the control objective is to drive the regulation error to zero (as opposed to driving the dynamic state T_{ai} to zero), the states of the system are redefined in terms of regulation error. The resulting augmented subsystem is defined as

$$\tilde{S}_i : \begin{bmatrix} \dot{\mathbf{e}}_i \\ \dot{z}_i \end{bmatrix} = \begin{bmatrix} \mathbf{A}_i & \mathbf{0}_{3 \times 1} \\ \mathbf{A}_{21} & \mathbf{0}_{1 \times 1} \end{bmatrix} \begin{bmatrix} \mathbf{e}_i \\ z_i \end{bmatrix} + \begin{bmatrix} \mathbf{B}_i \\ \mathbf{0}_{1 \times 2} \end{bmatrix} \mathbf{u}_i + \begin{bmatrix} \mathbf{A}_i \\ \mathbf{0}_{1 \times 3} \end{bmatrix} \mathbf{r}_{Li}, \quad (43)$$

where

$$\begin{aligned} \mathbf{e}_i &= \mathbf{x}_i - \mathbf{r}_{Li}, \\ \dot{z}_i &= T_{ai} - T_{ai,ref}, \\ \mathbf{A}_{21} &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}. \end{aligned} \quad (44)$$

The augmented subsystem is rewritten as

$$\tilde{S}_i : \dot{\tilde{\mathbf{x}}}_i = \tilde{\mathbf{A}}_i \tilde{\mathbf{x}}_i + \tilde{\mathbf{B}}_i \mathbf{u}_i + \tilde{\mathbf{D}}_i \mathbf{r}_{Li}. \quad (45)$$

With each of the subsystems being identical, the cost function for the local LQR formulation

$$J_i(\tilde{\mathbf{x}}_{i0}, \mathbf{u}_i) = \int_0^\infty (\tilde{\mathbf{x}}_i^T \mathbf{Q}_i \tilde{\mathbf{x}}_i + \mathbf{u}_i^T \mathbf{R}_i \mathbf{u}_i) dt, \quad (46)$$

can be applied to each subsystem, where $\tilde{\mathbf{x}}_{i0}$ is the initial state of each subsystem and

$$\begin{aligned}\mathbf{Q}_i &= \text{diag}(q_{i1}, 0, 0, q_{i4}), \\ \mathbf{R}_i &= \text{diag}(r_{i1}, r_{i2}),\end{aligned}\quad (47)$$

are chosen to penalize zone air temperature error and the integral of this error. Solving the LQ problem, the optimal control law is

$$\mathbf{u}_i = -\mathbf{K}_i \tilde{\mathbf{x}}_i, \quad (48)$$

where $\mathbf{K}_i \in \mathbb{R}^{2 \times 4}$.

Since the decentralized controller is designed using only the decoupled system representation, $\mathbf{S}_D$ (39), it is important to understand how this optimal control law performs when applied to the complete system representation, $\mathbf{S}$ (38), compared to a centralized controller designed with complete knowledge of the coupling. A centralized controller can be designed by solving the LQ problem for the complete system representation, $\mathbf{S}$ (38), using the centralized cost function

$$J(\tilde{\mathbf{x}}_0, \mathbf{u}) = \int_0^\infty (\tilde{\mathbf{x}}^T \mathbf{Q} \tilde{\mathbf{x}} + \mathbf{u}^T \mathbf{R} \mathbf{u}) dt, \quad (49)$$

where

$$\begin{aligned}\mathbf{Q} &= \text{diag}(\mathbf{Q}_1, \dots, \mathbf{Q}_n), \\ \mathbf{R} &= \text{diag}(\mathbf{R}_1, \dots, \mathbf{R}_n).\end{aligned}\quad (50)$$

The centralized control law

$$\mathbf{u} = -\mathbf{K} \tilde{\mathbf{x}}, \quad (51)$$

where $\mathbf{K} \in \mathbb{R}^{2n \times 4n}$, is used to determine the actuation of the n EEVs and n evaporator fans and has access to all of the states of each of the n evaporators.

From Definition 3.15 in [10], the decentralized control law is suboptimal with degree $\bar{\mu}$ if there exists a positive number $\bar{\mu}$ such that

$$J_D(\mathbf{x}_0) \leq \bar{\mu}^{-1} J(\mathbf{x}_0), \quad (52)$$

where $J(\mathbf{x}_0)$ is the cost associated with the centralized control law applied to the complete system, $\mathbf{S}$ (38), and $J_D(\mathbf{x}_0)$ is the cost associated with the decentralized control law applied to the same system. Using Theorem 3.18 in [10], the degree of suboptimality can be evaluated. A value of $\bar{\mu} = 1$ indicates that the decentralized control law in (48) has the same cost as the centralized control law in (51) when applied to the complete system. If $\bar{\mu} \ll 1$, then the cost of the decentralized control law may be significantly larger than the cost of the centralized control law, indicating degraded control performance when using decentralized control. In Section 5 we will evaluate $\bar{\mu}$ for the system considered in our case study.

4.2. Global control design

The objective of the global controller is to regulate the system to a desired value of composite superheat. Substituting (22) into (42) and simplifying gives

$$T_{SH} = T_{rq} - \frac{1}{n} \sum_{i=1}^n \eta_{ri} P_i. \quad (53)$$

Remark 4. The dynamic state T_{rq} evolves much faster than the dynamic states P_i , $i \in N$. Therefore, T_{rq} can be treated as an algebraic quantity, and we can redefine an augmented superheat as

$$\tilde{T}_{SH} = T_{SH} - T_{rq} = -\frac{1}{n} \sum_{i=1}^n \eta_{ri} P_i. \quad (54)$$

This claim will be verified in Section 5.1.2 in the context of the case study.

The parameter $\tilde{T}_{SH}$ can be expressed as an output of the following model:

$$\begin{aligned}\dot{\tilde{\mathbf{x}}}_G &= \underbrace{\begin{bmatrix} \mathbf{A}_F & \mathbf{0}_{(n+1) \times 1} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix}}_{\tilde{\mathbf{A}}} \tilde{\mathbf{x}}_G + \underbrace{\begin{bmatrix} \mathbf{B}_1 \\ \mathbf{B}_2 \end{bmatrix}}_{\tilde{\mathbf{B}}} \mathbf{u}_G \\ \tilde{\mathbf{y}}_G &= \tilde{\mathbf{C}} \tilde{\mathbf{x}}_G,\end{aligned}\quad (55)$$

where $\tilde{\mathbf{x}}_G = [\mathbf{x}_F \ T_{wc}]^T$, $\tilde{\mathbf{y}}_G = [\tilde{T}_{SH} \ P_2 \ \dots \ P_n \ P_c \ T_{wc}]^T$, $\mathbf{A}_F$ is given in (73), and

$$\begin{aligned}\mathbf{A}_{21} &= \begin{bmatrix} -\frac{\lambda_{rc} \beta_{k1} C_1}{C_q} & \dots & -\frac{\lambda_{rc} \beta_{kn} C_n}{C_q} & \frac{\eta_{rc} + \lambda_{rc} \beta_{k2}}{C_{wc} R_{rc}} - \frac{\lambda_{rc} \beta_{k1} C_c}{C_q} \end{bmatrix}, \mathbf{A}_{22} = -\frac{1}{C_{wc}} \left(\frac{1}{R_{rc}} + \frac{1}{R_{ac}} \right), \mathbf{B}_1 = \begin{bmatrix} \mathbf{0}_{n \times 1} & \mathbf{0}_{n \times 1} \\ \frac{\beta_{k3}}{C_c} & \mathbf{0}_{1 \times 1} \end{bmatrix}, \\ \mathbf{B}_2 &= \begin{bmatrix} \frac{\lambda_{rc} \beta_{k3}}{C_{wc} R_{rc}} & -\frac{K_{fc}}{C_{wc} R_{ac}} \end{bmatrix}, \tilde{\mathbf{C}} = \begin{bmatrix} C_1 \\ C_2 \end{bmatrix}, \mathbf{C}_1 = \begin{bmatrix} \frac{\eta_{r1}}{n} & \dots & \frac{\eta_{rn}}{n} & 0 & 0 \end{bmatrix}, \mathbf{C}_2 = [\mathbf{0}_{(n+1) \times 1} \ I_{(n+1) \times (n+1)}].\end{aligned}\quad (56)$$

Table 3

	Nominal	Minimum	Maximum	Units
Actuators				
ω_k	2500	2375	2625	RPM
ω_{fc}	1142	863	1421	RPM
a_{vi}	20	19	21	% open
ω_{fi}	897	632	1161	RPM
States				
P_i	330	307	355	kPa
P_c	1453	1340	1636	kPa
P_q	300	277	326	kPa
T_{wi}	9.5	6.9	11.4	°C
T_{ai}	14.9	13.1	16.1	°C
T_{wc}	48.9	46.0	53.4	°C
T_{rq}	21.6	13.7	23.7	°C
Other				
$T_{SH,i}$	18.9	6.3	25.6	°C
$\dot{Q}_{awi}$	0.93	0.64	1.37	kW
T_{ac}	30	–	–	°C

In the same way that each subsystem was augmented in (43), (55) is augmented such that the states represent the output regulation error and the integral of the regulation error for $\bar{T}_{SH}$. The augmented system is

$$\bar{\mathbf{S}}_G : \begin{bmatrix} \dot{\mathbf{e}}_G \\ \dot{\mathbf{z}}_G \end{bmatrix} = \begin{bmatrix} \tilde{\mathbf{C}}\tilde{\mathbf{A}}\tilde{\mathbf{C}}^{-1} & \mathbf{0}_{(n+2) \times 1} \\ \tilde{\mathbf{A}}_{21} & \mathbf{0}_{1 \times 1} \end{bmatrix} \begin{bmatrix} \mathbf{e}_i \\ \mathbf{z}_i \end{bmatrix} + \begin{bmatrix} \tilde{\mathbf{C}}\tilde{\mathbf{B}} \\ \mathbf{0}_{1 \times 2} \end{bmatrix} \mathbf{u}_i + \begin{bmatrix} \tilde{\mathbf{C}}\tilde{\mathbf{A}}\tilde{\mathbf{C}}^{-1} \\ \mathbf{0}_{1 \times (n+2)} \end{bmatrix} \mathbf{r}_G, \quad (57)$$

where $\mathbf{e}_G = \bar{\mathbf{y}}_G - \mathbf{r}_G$, $\dot{\mathbf{z}}_G = \bar{T}_{SH} - \bar{T}_{SH,ref}$, and $\tilde{\mathbf{A}}_{21} = \begin{bmatrix} 1 & \mathbf{0}_{(n-1) \times 1} & 0 & 0 \end{bmatrix}$. Eq. (57) can be written more compactly as

$$\bar{\mathbf{S}}_G : \dot{\bar{\mathbf{x}}}_G = \bar{\mathbf{A}}_G \bar{\mathbf{x}}_G + \bar{\mathbf{B}}_G \mathbf{u}_G + \bar{\mathbf{D}}_G \mathbf{r}_G. \quad (58)$$

The cost function for the global LQR formulation is

$$J_G(\bar{\mathbf{x}}_{G0}, \mathbf{u}_G) = \int_0^\infty (\bar{\mathbf{x}}_G^T \mathbf{Q}_G \bar{\mathbf{x}}_G + \mathbf{u}_G^T \mathbf{R}_G \mathbf{u}_G) dt, \quad (59)$$

where $\bar{\mathbf{x}}_{G0}$ is the initial state, and

$$\begin{aligned} \mathbf{Q}_G &= \text{diag}(q_1, \mathbf{q}_2, q_3, q_4, q_5), \\ \mathbf{R}_G &= \text{diag}(r_1, r_2), \end{aligned} \quad (60)$$

are chosen to penalize regulation error and the integral of regulation error for $\bar{T}_{SH}$ as well as T_{wc} . In choosing the values for q_1 through q_5 , q_1 is used to penalized $\bar{T}_{SH}$ regulation error, q_5 penalizes the integral of this error, $\mathbf{q}_2 = \mathbf{0}_{1 \times (n-1)}$ does not penalize changes in evaporator pressures P_2 through P_n , and $q_3 = 0$ does not penalize changes in the condenser pressure. The penalty on T_{wc} given by q_4 is used to capture the indirect influence of ω_{fc} on $\bar{T}_{SH}$. Solving the LQ problem, the optimal control law is

$$\mathbf{u}_G = -\mathbf{K}_G \bar{\mathbf{x}}_G, \quad (61)$$

where $\mathbf{K}_G \in \mathbb{R}^{2 \times (n+3)}$.

5. Case study

In this case study, the proposed control architecture is implemented on a 5-evaporator nonlinear VRF system model simulated using the Air Force Research Laboratory (AFRL) Transient Thermal Modeling and Optimization (ATTMO) toolbox [11] which is based on the Thermosys toolbox [12] from the University of Illinois at Urbana-Champaign (UIUC). The parameters of the linearized gray-box models of the fluid and thermal dynamics are identified based on this nonlinear model and presented in Table 4. Consistent with industrial practice, which modularizes the subsystems, it is assumed that all 5 evaporators, EEVs, and evaporator fans are identical to one another. A detailed description of the nonlinear 5-evaporator VRF system, as well as validation results of the parameter identification, are provided in the Appendix. As will be seen in the results of the case study, the controller successfully regulates to the desired setpoints despite the fact that both the evaporator fans and the EEVs are actuated beyond their respective linearized regions which are defined in the Appendix (see Table 3). This suggests that the modeling assumptions made earlier to derive a linearized model of the VRF dynamics are reasonable and suitable for the purpose of control-oriented modeling.

5.1. Numerical verification of assumptions

5.1.1. Fluid dynamics

Here we verify the assumptions stated in Remark 2 and Proposition 2 in the context of this case study. First we verify Remark 2.

Table 4
Identified fluid and thermal parameters.

Component	Fluid			Thermal		
Compressor	β_{k1} β_{k2} β_{k3}	1.218×10^{-4} 3.429×10^{-6} 1.170×10^{-5}	$[\text{kg (s kPa)}^{-1}]$ $[\text{kg (s kPa)}^{-1}]$ $[\text{kg (s RPM)}^{-1}]$	α_{k1}	2.090×10^{-2}	$[\text{kW } ^\circ\text{C}^{-1}]$
				α_{k2}	1.132×10^{-3}	$[\text{kW } ^\circ\text{C}^{-1}]$
				α_{k3}	3.115×10^{-4}	$[\text{kW RPM}^{-1}]$
				γ_{k1}	1.160×10^{-1}	$[\text{kW } ^\circ\text{C}^{-1}]$
				γ_{k2}	5.548×10^{-3}	$[\text{kW } ^\circ\text{C}^{-1}]$
				γ_{k3}	7.974×10^{-4}	$[\text{kW RPM}^{-1}]$
EEV	R_{vi} K_{vi}	4.042×10^5 4.785×10^1	$[\text{kPa s kg}^{-1}]$ $[\text{kPa (\% open)}^{-1}]$	R'_{vi}	6.102×10^1	$[\text{ } ^\circ\text{C kW}^{-1}]$
				K'_{vi}	6.941×10^{-1}	$[\text{ } ^\circ\text{C (\% open)}^{-1}]$
				K'_{ki}	5.805×10^{-3}	$[\text{ } ^\circ\text{C RPM}^{-1}]$
Condenser	C_c	4.725×10^{-4}	$[\text{kg kPa}^{-1}]$	C_{wc}	6.836×10^0	$[\text{kW s } ^\circ\text{C}^{-1}]$
				R_{ac}	2.675×10^0	$[\text{ } ^\circ\text{C kW}^{-1}]$
				R_{rc}	1.328×10^0	$[\text{ } ^\circ\text{C kW}^{-1}]$
				η_{rc}	2.599×10^{-2}	$[\text{ } ^\circ\text{C kPa}^{-1}]$
				K_{fc}	1.019×10^{-2}	$[\text{ } ^\circ\text{C RPM}^{-1}]$
				λ_{rc}	1.491×10^2	$[\text{ } ^\circ\text{C s kg}^{-1}]$
Evaporator	C_i	1.103×10^{-4}	$[\text{kg kPa}^{-1}]$	C_{wi}	9.917×10^{-1}	$[\text{kW s } ^\circ\text{C}^{-1}]$
				R_{ai}	8.359×10^0	$[\text{ } ^\circ\text{C kW}^{-1}]$
				R_{ri}	1.071×10^1	$[\text{ } ^\circ\text{C kW}^{-1}]$
				η_{ri}	1.898×10^{-2}	$[\text{ } ^\circ\text{C kPa}^{-1}]$
				K_{fi}	7.428×10^{-3}	$[\text{ } ^\circ\text{C RPM}^{-1}]$
				λ_{ri}	1.347×10^3	$[\text{ } ^\circ\text{C s kg}^{-1}]$
Pipe	R_{qi}	1.093×10^4	$[\text{kPa s kg}^{-1}]$	μ_{m1}	7.350×10^1	$[\text{ } ^\circ\text{C kW}^{-1}]$
				μ_{m2}	1.895×10^4	$[\text{ } ^\circ\text{C s kg}^{-1}]$
				μ_{m3}	4.496×10^1	$[\text{ } ^\circ\text{C kW}^{-1}]$
				μ_{m4}	1.208×10^4	$[\text{ } ^\circ\text{C s kg}^{-1}]$
Junction	C_q	5.183×10^{-4}	$[\text{kg kPa}^{-1}]$	C_{rq}	2.312×10^{-2}	$[\text{kW s } ^\circ\text{C}^{-1}]$
				κ_{rq}	3.800×10^{-3}	$[\text{kW s kg}^{-1}]$

The matrix $\mathbf{A}_F$ describing the fluid dynamics of the 5-evaporator VRF system is given by

$$\mathbf{A}_F \mathbf{x}_F = \begin{bmatrix} -1.03 & -0.177 & -0.177 & -0.177 & -0.177 & -0.734 \\ -0.177 & -1.03 & -0.177 & -0.177 & -0.177 & -0.734 \\ -0.177 & -0.177 & -1.03 & -0.177 & -0.177 & -0.734 \\ -0.177 & -0.177 & -0.177 & -1.03 & -0.177 & -0.734 \\ -0.177 & -0.177 & -0.177 & -0.177 & -1.03 & -0.734 \\ -0.0496 & -0.0496 & -0.0496 & -0.0496 & -0.0496 & -0.254 \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \\ P_5 \\ P_c \end{bmatrix}. \quad (62)$$

The eigenvalues of $\mathbf{A}_F$ are -1.85 , -0.852 , -0.852 , -0.852 , -0.852 , and -0.140 . We will now argue that the slowest eigenvalue, -0.140 , corresponds to the dynamics of P_c . Define $\lambda = -0.140$, and let $\mathbf{w}^T$ be the left eigenvector of $\mathbf{A}_F$ corresponding to the eigenvalue λ ; for the above matrix, we have

$$\mathbf{w}^T = [-0.0310 \quad -0.0310 \quad -0.0310 \quad -0.0310 \quad -0.0310 \quad 0.998]. \quad (63)$$

Define the similarity transformation matrix

$$\mathbf{U} = \begin{bmatrix} I_{n \times n} & \mathbf{0}_{n \times 1} \\ & \mathbf{w}^T \end{bmatrix}, \quad (64)$$

and the new state vector

$$\tilde{\mathbf{x}} = \begin{bmatrix} \tilde{x}_1 \\ \vdots \\ \tilde{x}_{n+1} \end{bmatrix} = \mathbf{U} \mathbf{x}_F. \quad (65)$$

This gives us $\tilde{x}_i = P_i$ for $i \in N$ and $\tilde{x}_{n+1} = \mathbf{w}^T \mathbf{x}_F$. Based on $\mathbf{w}^T$ given above, we see that $\tilde{x}_{n+1}(t) = -0.0310(P_1(t) + P_2(t) + P_3(t) + P_4(t) + P_5(t)) + 0.998P_c(t)$. When the magnitude of $P_c(t)$ is of the same order (or larger than) the magnitudes of the other pressures in the system, the fact that the coefficient of P_c is approximately 32.2 times larger than the coefficient of the other pressures indicates that $\tilde{x}_{n+1}(t) \approx P_c(t)$. Furthermore, we have $\dot{\tilde{x}}_{n+1} = \mathbf{w}^T \dot{\mathbf{x}}_F = \mathbf{w}^T \mathbf{A}_F \mathbf{U}^{-1} \tilde{\mathbf{x}} = \lambda \mathbf{w}^T \mathbf{U}^{-1} \tilde{\mathbf{x}} = \lambda \tilde{x}_{n+1}$, and thus the dynamics of P_c correspond to the slowest eigenvalue -0.140 . The physical intuition for this result comes from the fact that the fluid capacitance of the condenser, $C_c = 4.72 \times 10^{-4} \text{ kg kPa}^{-1}$ is 4 times larger than the fluid capacitance of each of the five evaporators, $C_1 = C_2 = \dots = C_5 = 1.10 \times 10^{-4} \text{ kg kPa}^{-1}$. Consequently, we will treat the dynamic effect of P_c on P_1 through P_5 as a slowly varying disturbance and will show later that our controller designed under this assumption works well in practice. Moreover, the dynamics of P_1 through P_5 will reach steady-state much more quickly than those of P_c .

Therefore, a global controller designed with feedback of P_c can be used to meet global control objectives without being affected by the effect of the local controllers on the individual evaporator pressures.

Now we verify the claim made in [Proposition 2](#). The reader can easily verify that the 5×5 block of the matrix $\mathbf{A}_F$, shown in (66), is indeed diagonally dominant. Therefore the coupling between the individual evaporator pressures is weak and the dynamics lend themselves to a decentralized control architecture at the local level [\[10\]](#).

$$\mathbf{A}_{F,5 \times 5} = \begin{bmatrix} -1.03 & -0.177 & -0.177 & -0.177 & -0.177 \\ -0.177 & -1.03 & -0.177 & -0.177 & -0.177 \\ -0.177 & -0.177 & -1.03 & -0.177 & -0.177 \\ -0.177 & -0.177 & -0.177 & -1.03 & -0.177 \\ -0.177 & -0.177 & -0.177 & -0.177 & -1.03 \end{bmatrix} \quad (66)$$

5.1.2. Thermal dynamics

Here we verify the assumption made in [Remark 4](#) in Section 4.2 stating that T_{rq} can be treated as an algebraic quantity. Based on the model presented in Section 2.2, $\hat{T}_{rq}$ is a function of its own state T_{rq} and the fluid dynamic states, P_c and P_i for $i \in N$. The matrix $\mathbf{A}_F$ can be augmented with T_{rq} and is given by

$$\hat{\mathbf{A}}_F \hat{\mathbf{x}}_F = \begin{bmatrix} -1.03 & -0.177 & -0.177 & -0.177 & -0.177 & -0.734 & 0.00 \\ -0.177 & -1.03 & -0.177 & -0.177 & -0.177 & -0.734 & 0.00 \\ -0.177 & -0.177 & -1.03 & -0.177 & -0.177 & -0.734 & 0.00 \\ -0.177 & -0.177 & -0.177 & -1.03 & -0.177 & -0.734 & 0.00 \\ -0.177 & -0.177 & -0.177 & -0.177 & -1.03 & -0.734 & 0.00 \\ -0.0496 & -0.0496 & -0.0496 & -0.0496 & -0.0496 & -0.254 & 0.00 \\ 9.44 \times 10^3 & 9.44 \times 10^3 & 9.44 \times 10^3 & 9.44 \times 10^3 & 9.44 \times 10^3 & 4.04 \times 10^4 & -2.35 \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \\ P_5 \\ P_c \\ T_{rq} \end{bmatrix}, \quad (67)$$

where the eigenvalues of $\mathbf{A}_F$ are $-1.85, -0.852, -0.852, -0.852, -0.852, -0.140$, and -2.35 . A comparison against the eigenvalues of $\mathbf{A}_F$ indicates that the fastest eigenvalue, $\{-2.35\}$, corresponds to the dynamics of T_{rq} . Given that $\tilde{T}_{SH}$ (54) is defined as a linear combination of P_1 through P_5 (whose dynamics correspond to the eigenvalues $-1.85, -0.852, -0.852, -0.852, -0.852$), T_{rq} will evolve at least 2.76 times as fast as $\tilde{T}_{SH}$, thereby validating our decision to regulate $\tilde{T}_{SH}$ as opposed to T_{SH} .

5.2. Controller design

To obtain the local controller gain matrix for each of the 5 local controllers, the LQ problem defined in (46) is solved using

$$\begin{aligned} \mathbf{Q}_i &= \text{diag}(1, 0, 0, 10), \\ \mathbf{R}_i &= \text{diag}(0.3, 250), \end{aligned} \quad (68)$$

which results in the local controller gain matrix

$$\mathbf{K}_i = - \begin{bmatrix} 9.96 \times 10^2 & 8.36 \times 10^0 & 2.05 \times 10^{-2} & 5.03 \times 10^0 \\ 2.02 \times 10^1 & 1.70 \times 10^{-1} & 4.18 \times 10^{-4} & 9.81 \times 10^{-2} \end{bmatrix}, \quad i \in N. \quad (69)$$

Using Theorem 3.18 in [\[10\]](#), the degree of suboptimality was calculated to be $\bar{\mu} = 0.988$, resulting in $J_D(\mathbf{x}_0) \leq 1.01J(\mathbf{x}_0)$. This indicates that the cost associated with our decentralized controllers will not exceed the optimal cost for a centralized controller by more than 1%.

To obtain the global controller gain matrix $\mathbf{K}_G$, the LQ problem defined in (59) is solved using

$$\begin{aligned} \mathbf{Q}_G &= \text{diag}(10, 0_{1 \times (n-1)}, 0, 1, 0.1), \\ \mathbf{R}_G &= \text{diag}(6 \times 10^{-2}, 2.5 \times 10^{-5}), \end{aligned} \quad (70)$$

which results in

$$\mathbf{K}_G = \begin{bmatrix} -1.28 & 0 & 0 & 0 & 0 & 0.0704 & -0.0110 & -1.29 \\ 25.1 & 0 & 0 & 0 & 0 & 1.07 & -61.3 & 0.0957 \end{bmatrix}. \quad (71)$$

Note that the 0 elements in $\mathbf{K}_G$ indicate that the global actuators (ω_k and ω_{fc}) do not directly affect the individual evaporator pressures; however, the effect of the pressure dynamics is captured indirectly through $\tilde{T}_{SH}$ which is defined as a weighted average of the evaporator pressures as shown in (54).

5.3. Simulated results

The efficacy of the proposed control architecture for ensuring that the control objectives of the system are achieved is demonstrated through the following scenarios simulated using the ATTMO toolbox for MATLAB Simulink. In Scenario 1, we show that in the *absence* of the global controller, the saturation of evaporator fans can prevent the local controllers from meeting their respective performance objectives and also lead to a loss of composite superheat (at which point the system is considered inoperable due to the potential damage caused by liquid entering the compressor). In Scenario 2, we quantify how the desired value of composite superheat directly affects total energy

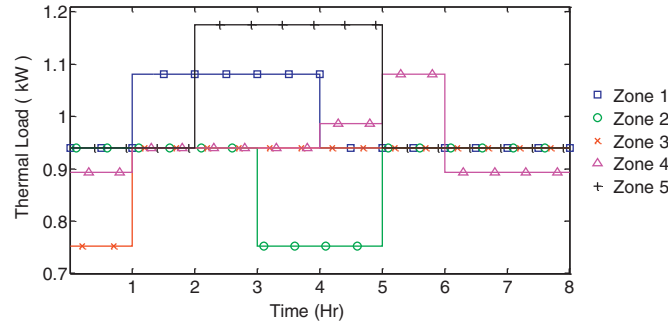


Fig. 8. Thermal load disturbances $\dot{Q}_{Li}$, $i \in N$ acting on each evaporator over the 8 h simulation period.

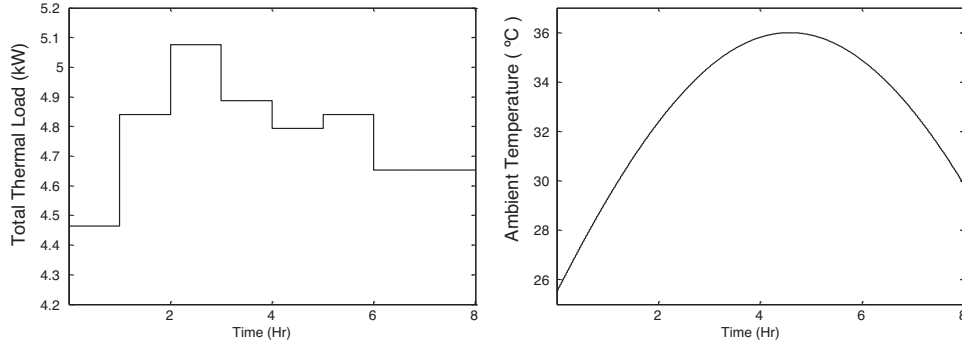


Fig. 9. Total thermal load on system (left) and ambient temperature (T_{ac}) disturbance (right).

consumption through a comparison of composite superheat regulation to several different values using the partially decentralized control architecture.

In each simulated test case, the simulated nonlinear VRF system is perturbed by lumped load disturbances for each of the 5 zones over an 8 h time period, as shown in Fig. 8. The magnitude of the load disturbance in each zone is designed to be between 5% and 25% of the nominal cooling load of 0.94 kW. The desired zone air temperatures for each of the 5 zones are 20 °C, 20 °C, 21 °C, 21 °C, and 21.5 °C, respectively, and are held constant over the 8 h simulation period. The system is also perturbed by a time-varying ambient temperature disturbance, shown in Fig. 9.

5.3.1. Scenario 1: Achieving local and global control objectives

First, only the n local controllers are implemented to demonstrate the necessity of the global controller in achieving the local objectives. Each local controller must regulate its local zone air temperature to a constant reference, $T_{ai,ref}$, while rejecting a time-varying cooling load disturbance, $\dot{Q}_{Li}$, with the compressor and condenser fan operating at fixed values of $\omega_k = 1791$ rpm and $\omega_{fc} = 1097$ rpm, respectively. The local actuators are bounded above and below by the following limits: $0 \text{ rpm} < \omega_{fi} \leq 1600$ rpm and $0\% < a_{vi} \leq 100\%$ for all $i \in N$.

In response to the first step increase in the thermal load in zones 1, 4, and 5 at $t = 1$ h (see Fig. 9), the evaporator fan speeds and EEV apertures increase in an effort to increase the amount of cooling being delivered to each of the zones. However, the opening of the EEVs causes the evaporator superheat in each EEV to decrease (Fig. 12). The evaporator fans and EEVs ultimately saturate at their upper limits, causing large regulation errors for the zone air temperatures and complete loss of evaporator superheat (Figs. 10–13). At this point, the system is considered inoperable to prevent damage to the compressor. By operating the compressor and condenser fan at fixed speeds, the system was forced to operate with a limited refrigerant flow rate, as well as a limited amount of heat rejection capability through the condenser, which was insufficient for the increase in thermal loads in each of the zones.

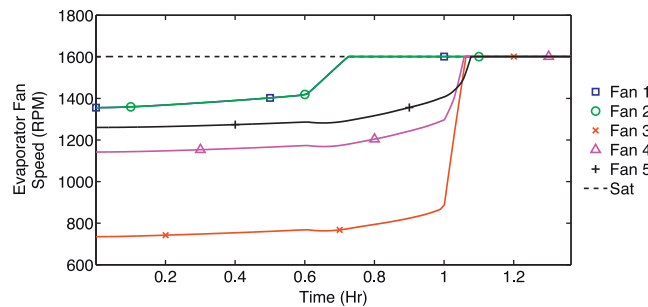


Fig. 10. In the absence of composite superheat regulation, the 5 evaporator fan control signals saturate at a value of 1600 rpm shortly after the step change in thermal load disturbances at $t = 1$ h.

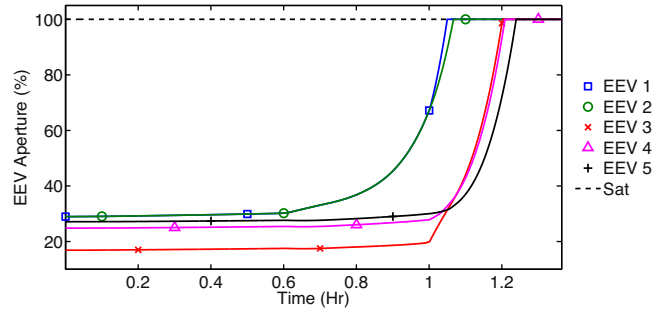


Fig. 11. In the absence of composite superheat regulation, the 5 EEV control signals saturate at an aperture of 100% shortly after the evaporator fan control signals saturate at their upper bound (see Fig. 10).

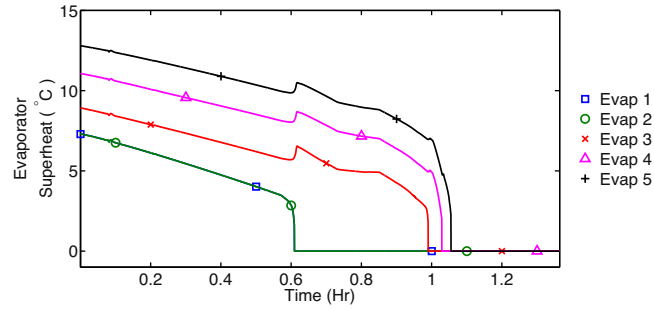


Fig. 12. In the absence of composite superheat regulation, each evaporator experiences a loss of superheat.

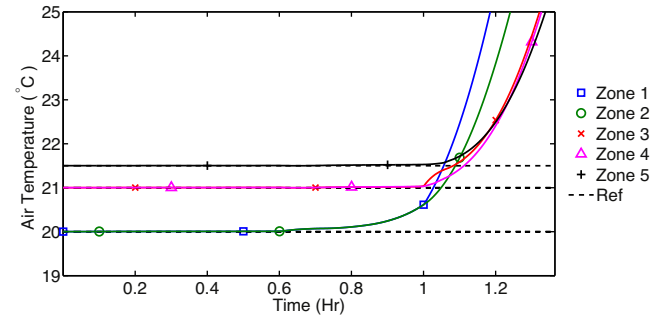


Fig. 13. Due to the saturation of both the evaporator fans and the EEVs (see Figs. 10 and 11), the local controllers are unable to reject the thermal load disturbances and regulate zone air temperatures T_{a1} through T_{a5} to their desired values.

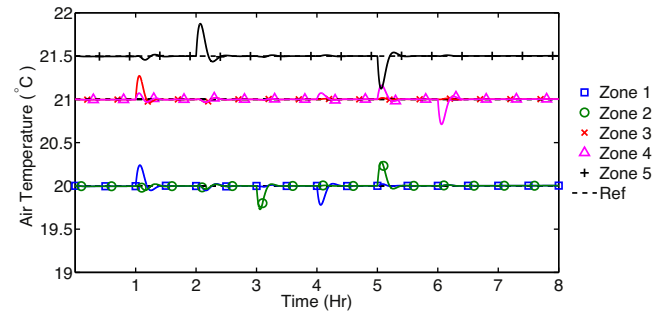


Fig. 14. With the global controller regulating composite superheat (see Fig. 17), each local controller is able to regulate its corresponding zone air temperature.

Next, the global controller is introduced into the system to regulate composite superheat to 10°C. The performance of the local and global controllers is shown in Figs. 14 and 17, respectively. The local control input signals are shown in Figs. 15 and 16, and the global control input signals are shown in Fig. 18.

Although evaporator fans 1, 4, and 5 saturate for varying durations over the 8 h simulation period as shown in Fig. 15, the local control objectives are met. This is due to the increase in compressor speed shown in Fig. 18 which mirrors the trajectory of the total thermal load on the system (shown in Fig. 9).

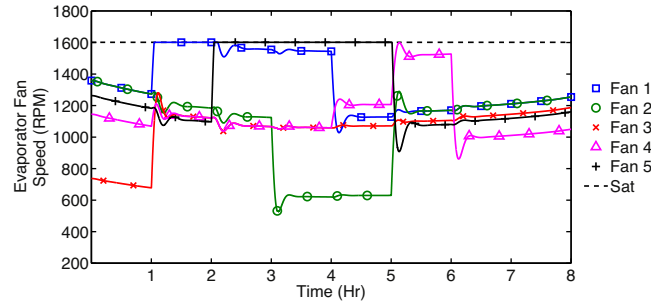


Fig. 15. Due to the regulation of composite superheat, local performance objectives are met (see Fig. 14) despite the saturation of evaporator fans 1 and 5.

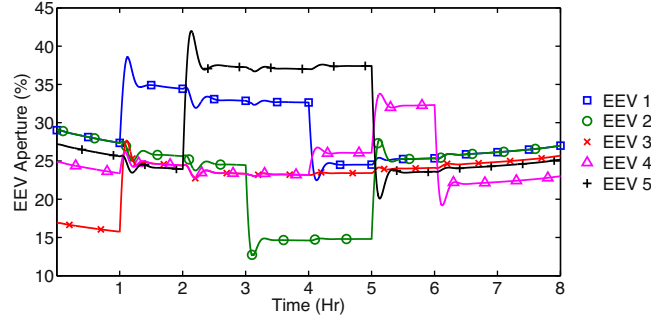


Fig. 16. Due to the regulation of composite superheat, the EEV control input signals stay well within their saturation limits for the duration of the 8 h simulation period.

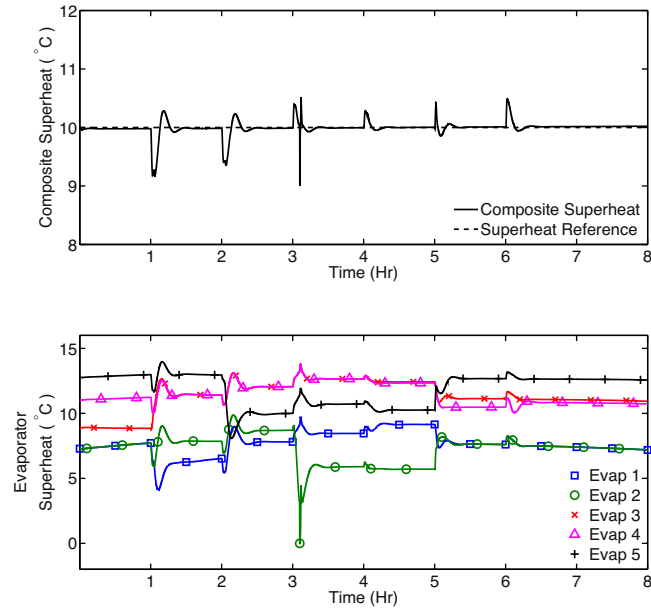


Fig. 17. Performance of composite superheat (top) and individual evaporator superheat signals (bottom).

The global controller is able to regulate composite superheat within roughly 1°C of the reference of 10°C , as shown in Fig. 17. Despite a brief loss of superheat in Evaporator 2 at $t = 3$ h, composite superheat was easily regulated to 10°C , thereby ensuring that only superheated vapor entered the compressor.

5.3.2. Scenario 2: Achieving the global control objectives

As discussed in Section 3, the global objective of composite superheat regulation is defined to both ensure safe compressor operation and reduce total energy consumption. The importance of composite superheat regulation in relation to achieving local performance objectives was highlighted in the previous section. Here we characterize the relationship between composite superheat regulation and total energy consumption.

It is well established in the VCS literature that lower evaporator superheat leads to higher operational efficiency [9]. Using the same local and global controllers, the same 8 h simulation test was repeated for 8 different desired values of composite superheat. The total energy consumption resulting from each test is compared in Fig. 20. To ensure a fair comparison between tests, the composite superheat tracking performance of the global controller for each test was characterized using the root-mean-square (RMS) error between $T_{SH,ref}$ and

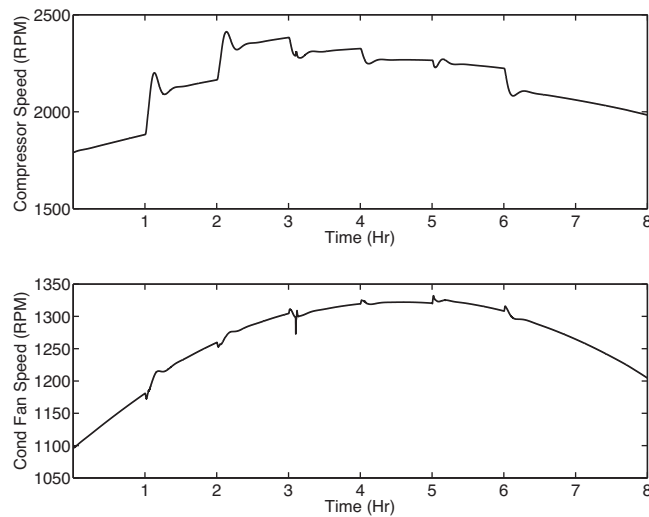


Fig. 18. Global control input signals: compressor speed, ω_k (top) and condenser fan speed, ω_{fc} (bottom); the shape of ω_k and ω_{fc} mirrors that of the total thermal load on the system shown in Fig. 9.

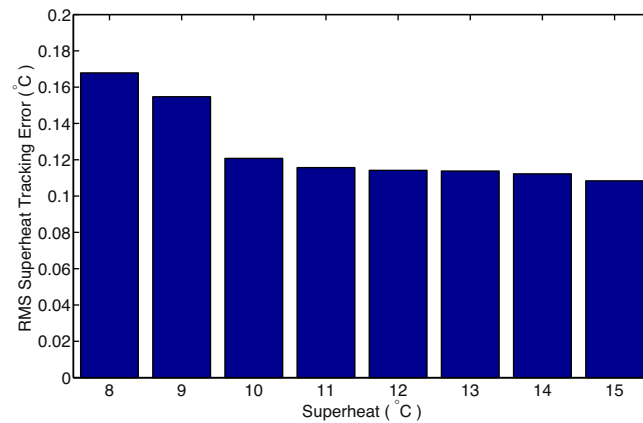


Fig. 19. RMS composite superheat regulation error for different composite superheat reference values.

T_{SH} . Fig. 19 shows that the maximum difference in RMS error between any two tests is 0.060°C which is well within the accuracy of a typical thermocouple. Therefore, the same global controller was able to regulate the different desired composite superheat values with nearly equal performance.

From Fig. 20 it appears that the relative total energy consumption grows with increasing values of superheat. A lower value of composite superheat clearly reduces energy consumption; however, for decreasing values of T_{SH} , the incremental energy savings diminish. This emphasizes the novelty of the global control objective defined here, which is simply the regulation of a single quantity – composite

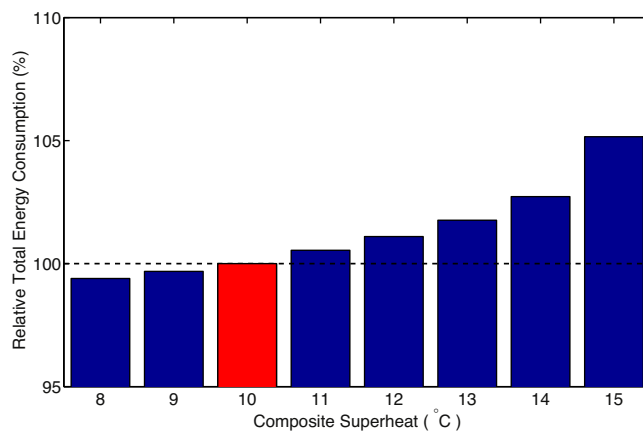


Fig. 20. Relative total energy consumption resulting from regulation of composite superheat to different reference values; reference case of $T_{SH,ref} = 10^\circ\text{C}$ is highlighted in red.

superheat – which simultaneously addresses system performance objectives and reduces total energy consumption. Moreover, in the partially decentralized control architecture, there is no communication between the local and global controllers. Only the n evaporator pressures need to be measured by the global controllers, thereby addressing our desire to design an architecture which is easily implementable from the viewpoint of industrial practitioners.

6. Conclusion

In this paper, we presented a scalable modeling framework for variable-refrigerant-flow (VRF) systems. The gray-box modeling structure for the fluid and thermal dynamics provided insights into the natural dynamics of these highly complex systems. Using this new understanding of the underlying dynamics, a partially decentralized control architecture was proposed to meet both local and global objectives for the system. The decentralization of the local controllers allows this architecture to be easily scaled to systems with $n \gg 2$ evaporators. The limited one-way sensor information flow from the individual evaporators to the global controller makes this control architecture amenable to industrial implementation while still providing a high level of performance and a reduction in total energy consumption. Future work will include a robustness analysis of the proposed control architecture to faults in the system as well as experimental validation. Additionally, this framework will be applied to systems with different components (including non-identical evaporators) to investigate the generalizability of this approach to a wide range of systems.

$$\begin{aligned}
 \begin{bmatrix} \dot{P}_1 \\ \dot{P}_2 \\ \vdots \\ \dot{P}_n \\ \dot{P}_c \\ \dot{P}_q \end{bmatrix} &= \begin{bmatrix} -\frac{1}{C_1} \left(\frac{1}{R_{v1}} + \frac{1}{R_{q1}} \right) & 0 & \dots & 0 & \frac{1}{C_1 R_{v1}} & \frac{1}{C_1 R_{q1}} \\ 0 & -\frac{1}{C_2} \left(\frac{1}{R_{v2}} + \frac{1}{R_{q2}} \right) & \ddots & \vdots & \frac{1}{C_2 R_{v2}} & \frac{1}{C_2 R_{q2}} \\ \vdots & \ddots & \ddots & 0 & \vdots & \vdots \\ 0 & \dots & 0 & -\frac{1}{C_n} \left(\frac{1}{R_{vn}} + \frac{1}{R_{qn}} \right) & \frac{1}{C_n R_{vn}} & \frac{1}{C_n R_{qn}} \\ \hline \frac{1}{C_c R_{v1}} & \frac{1}{C_c R_{v2}} & \dots & \frac{1}{C_c R_{vn}} & \frac{1}{C_c} \left(\beta_{k2} - \sum_{i=1}^n \frac{1}{R_{vi}} \right) & \frac{\beta_{k1}}{C_c} \\ \frac{1}{C_q R_{q1}} & \frac{1}{C_q R_{q2}} & \dots & \frac{1}{C_q R_{qn}} & -\frac{\beta_{k2}}{C_q} & -\frac{1}{C_q} \left(\beta_{k1} + \sum_{i=1}^n \frac{1}{R_{qi}} \right) \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ \vdots \\ P_n \\ P_c \\ P_q \end{bmatrix} \\
 &+ \begin{bmatrix} \frac{K_{v1}}{C_1 R_{v1}} & 0 & \dots & 0 & 0 \\ 0 & \frac{K_{v2}}{C_2 R_{v2}} & \ddots & \vdots & 0 \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \dots & 0 & \frac{K_{vn}}{C_n R_{vn}} & 0 \\ \hline -\frac{K_{v1}}{C_c R_{v1}} & -\frac{K_{v2}}{C_c R_{v2}} & \dots & -\frac{K_{vn}}{C_c R_{vn}} & \frac{\beta_{k3}}{C_c} \\ 0 & 0 & \dots & 0 & -\frac{\beta_{k3}}{C_q} \end{bmatrix} \begin{bmatrix} a_{v1} \\ a_{v2} \\ \vdots \\ a_{vn} \\ \omega_k \end{bmatrix}
 \end{aligned} \tag{72}$$

$$\begin{aligned}
 \mathbf{A}_F &= \begin{bmatrix} -\frac{1}{C_1} \left(\frac{1}{R_{v1}} + \frac{1}{R_{q1}} \right) - \frac{1}{C_q R_{q1}} & -\frac{C_2}{C_q C_1 R_{q1}} & \dots & -\frac{C_n}{C_q C_1 R_{q1}} & \frac{1}{C_1} \left(\frac{1}{R_{v1}} - \frac{C_c}{C_q R_{q1}} \right) \\ -\frac{C_1}{C_q C_2 R_{q2}} & -\frac{1}{C_2} \left(\frac{1}{R_{v2}} + \frac{1}{R_{q2}} \right) - \frac{1}{C_q R_{q2}} & \ddots & \vdots & \frac{1}{C_2} \left(\frac{1}{R_{v2}} - \frac{C_c}{C_q R_{q2}} \right) \\ \vdots & \ddots & \ddots & -\frac{C_n}{C_q C_{n-1} R_{q(n-1)}} & \vdots \\ -\frac{C_1}{C_q C_n R_{qn}} & \dots & -\frac{C_{n-1}}{C_q C_n R_{qn}} & -\frac{1}{C_n} \left(\frac{1}{R_{vn}} + \frac{1}{R_{qn}} \right) - \frac{1}{C_q R_{qn}} & \frac{1}{C_n} \left(\frac{1}{R_{vn}} - \frac{C_c}{C_q R_{qn}} \right) \\ \hline \frac{1}{C_c} \left(\frac{1}{R_{v1}} - \frac{\beta_{k1} C_1}{C_q} \right) & \frac{1}{C_c} \left(\frac{1}{R_{v2}} - \frac{\beta_{k1} C_2}{C_q} \right) & \dots & \frac{1}{C_c} \left(\frac{1}{R_{vn}} - \frac{\beta_{k1} C_n}{C_q} \right) & \frac{1}{C_c} \left(\beta_{k2} - \sum_{i=1}^n \frac{1}{R_{vi}} - \frac{\beta_{k1} C_c}{C_q} \right) \end{bmatrix} \\
 \mathbf{B}_F &= \begin{bmatrix} \frac{K_{v1}}{C_1 R_{v1}} & 0 & \dots & 0 & 0 \\ 0 & \frac{K_{v2}}{C_2 R_{v2}} & \ddots & \vdots & 0 \\ \vdots & \ddots & \ddots & 0 & \vdots \\ 0 & \dots & 0 & \frac{K_{vn}}{C_n R_{vn}} & 0 \\ \hline -\frac{K_{v1}}{C_c R_{v1}} & -\frac{K_{v2}}{C_c R_{v2}} & \dots & -\frac{K_{vn}}{C_c R_{vn}} & \frac{\beta_{k3}}{C_c} \end{bmatrix}
 \end{aligned} \tag{73}$$

$$\begin{aligned}
\mathbf{A}_T &= \begin{bmatrix} -\frac{1}{C_{a1}R_{a1}} & \frac{1}{C_{a1}R_{a1}} & & & & \\ \frac{1}{C_{w1}R_{w1}} & -\frac{1}{C_{w1}}\left(\frac{1}{R_{w1}} + \frac{1}{R_{r1}}\right) & & & & \\ & & -\frac{1}{C_{a2}R_{a2}} & \frac{1}{C_{a2}R_{a2}} & & \\ & & \frac{1}{C_{w2}R_{w2}} & -\frac{1}{C_{w2}}\left(\frac{1}{R_{w2}} + \frac{1}{R_{r2}}\right) & & \\ & & & & \ddots & \\ & & & & & 0_{2 \times 2} \\ & & & & & -\frac{1}{C_{an}R_{an}} & \frac{1}{C_{an}R_{an}} \\ & & & & & \frac{1}{C_{wn}R_{wn}} & -\frac{1}{C_{wn}}\left(\frac{1}{R_{wn}} + \frac{1}{R_{rn}}\right) \end{bmatrix} \\
\mathbf{B}_T &= \begin{bmatrix} -\frac{K_{f1}}{C_{a1}R_{a1}} & 0 & & & & 0_{2 \times 2} \\ \frac{K_{f1}}{C_{w1}R_{w1}} & -\frac{K_{r1}\lambda_{r1}}{C_{w1}R_{r1}R_{v1}} & & & & \\ & & -\frac{K_{f2}}{C_{a2}R_{a2}} & 0 & & \\ & & \frac{K_{f2}}{C_{w2}R_{w2}} & -\frac{K_{r2}\lambda_{r2}}{C_{w2}R_{r2}R_{v2}} & & \\ & & & & \ddots & \\ & & & & & 0_{2 \times 2} \\ & & & & & -\frac{K_{fn}}{C_{an}R_{an}} & 0 \\ & & & & & \frac{K_{fn}}{C_{wn}R_{wn}} & -\frac{K_{rn}\lambda_{rn}}{C_{wn}R_{rn}R_{vn}} \end{bmatrix} \\
[\mathbf{w}_{r1} \quad \mathbf{w}_{r2}] &= \left[\begin{array}{cccccc} \frac{1}{C_{a1}} & 0 & & & & 0_{2 \times 2} \\ 0 & \frac{\eta_{r1}}{C_{w1}R_{w1}} + \frac{\lambda_{r1}}{C_{w1}R_{r1}R_{v1}} & & & & \\ & & \frac{1}{C_{a2}} & 0 & & \\ & & 0 & \frac{\eta_{r2}}{C_{w2}R_{w2}} + \frac{\lambda_{r2}}{C_{w2}R_{r2}R_{v2}} & & \\ & & & & \ddots & \\ & & & & & 0_{2 \times 2} \\ & & & & & \frac{1}{C_{an}} & 0 \\ & & & & & 0 & \frac{\eta_{rn}}{C_{wn}R_{wn}} + \frac{\lambda_{rn}}{C_{wn}R_{rn}R_{vn}} \end{array} \right] \left[\begin{array}{c} 0 \\ -\frac{\lambda_{r1}}{C_{w1}R_{r1}R_{v1}} \\ 0 \\ -\frac{\lambda_{r2}}{C_{w2}R_{r2}R_{v2}} \\ \vdots \\ 0 \\ -\frac{\lambda_{rn}}{C_{wn}R_{rn}R_{vn}} \end{array} \right]
\end{aligned} \quad (74)$$

Appendix A.

The gray-box modeling approach described in Section 2 requires the following assumptions:

- all of the fluid and thermal dynamics are linear within a neighborhood of a nominal operating condition
- the electrical analogies in Tables 1 and 2 are valid
- $C_i \ll C_q$ for all systems
- $C_c \gg C_i$ for all systems
- the temperature of the metal coil of each evaporator can be modeled as a single lumped temperature
- the temperature of the refrigerant in each evaporator can be modeled using a single lumped temperature which is directly proportional to the associated pressure of the refrigerant
- zones have a single lumped capacitance and temperature and are not thermally coupled to one another
- the fluid exiting each of the evaporators is single-phase (superheated vapor) but the majority of each evaporator is filled with two-phase fluid.

These assumptions are validated through the case study presented in Section 5. Here we provide details about the nonlinear VRF system model and the parameter identification of the linear gray-box model which are used in the case study. The 5-evaporator VRF system was

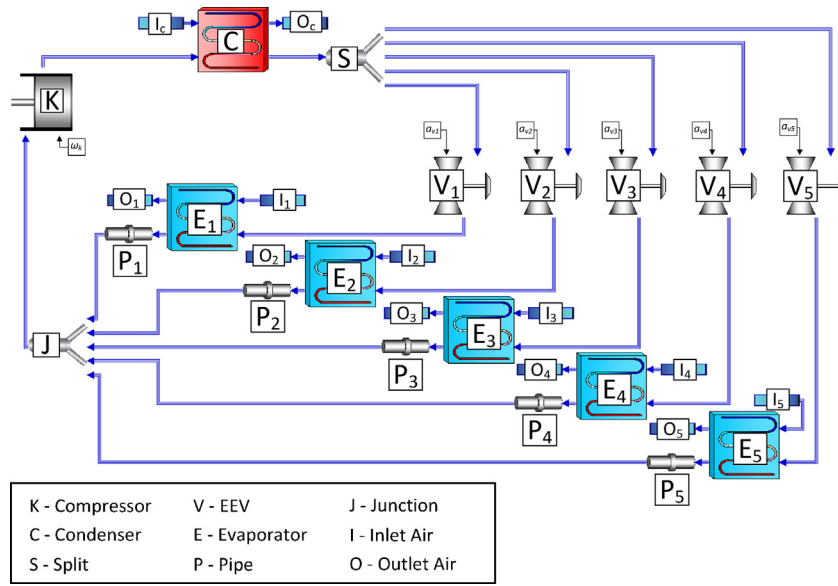


Fig. 21. ATTMO model of 5-evaporator system used in the case study. Note that the Inlet Air represents the zone air temperature regulated by the local controllers.

simulated using the Air Force Research Laboratory (AFRL) Transient Thermal Modeling and Optimization (ATTMO) toolbox [11] and is shown in Fig. 21. It is assumed that all 5 evaporators, EEVs, and evaporator fans are identical to one another. Each of the heat exchangers are of the coil-and-fin configuration, and the evaporators are modeled after the evaporators for the physical dual-evaporator system in [12] while the compressor and condenser were scaled appropriately.

To identify the parameters used in (73) and (74) to represent the fluid and thermal dynamics, respectively, pseudo-random binary signals (PRBSs) were sent to each of the actuators. The nominal values about which the system was perturbed, along with their range, are given in Table 3. A least-squares based approach was used to identify each of the parameters using the input and corresponding output signals. The identified parameters are shown in Table 3. Note that standard SI units are used in this paper and are as follows: pressure (kPa), mass flow rate (kg s^{-1}), temperature ($^{\circ}\text{C}$), and heat transfer rate (kW). Representative model validation plots for a selection of the fluid parameters are shown in Fig. 22, where the 3 subplots correspond to the validation of (2), (4), and (6), respectively. Similarly, Fig. 23 shows the validation of a selection of the thermal parameters, where the 4 subplots correspond to the validation of (15), (18), (22), and (21).

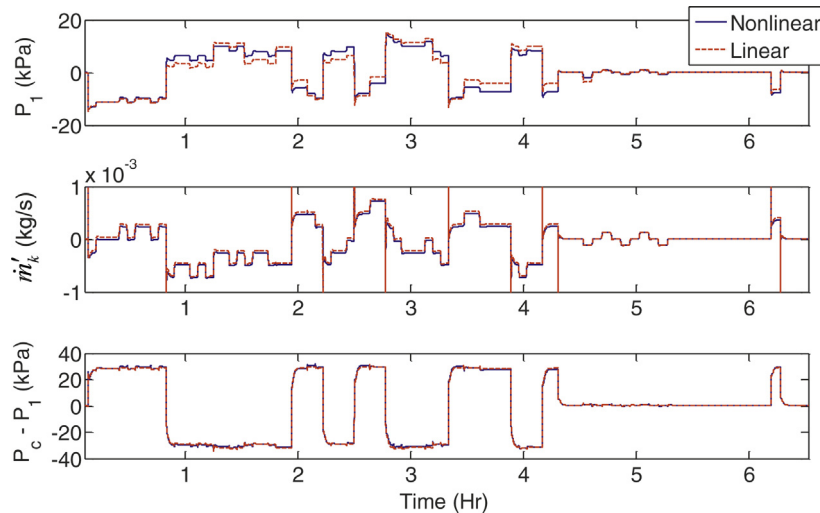


Fig. 22. Parameter identification validation for the fluid model through a comparison of the ATTMO nonlinear model and the linearized fluid dynamics model presented in Section 2.

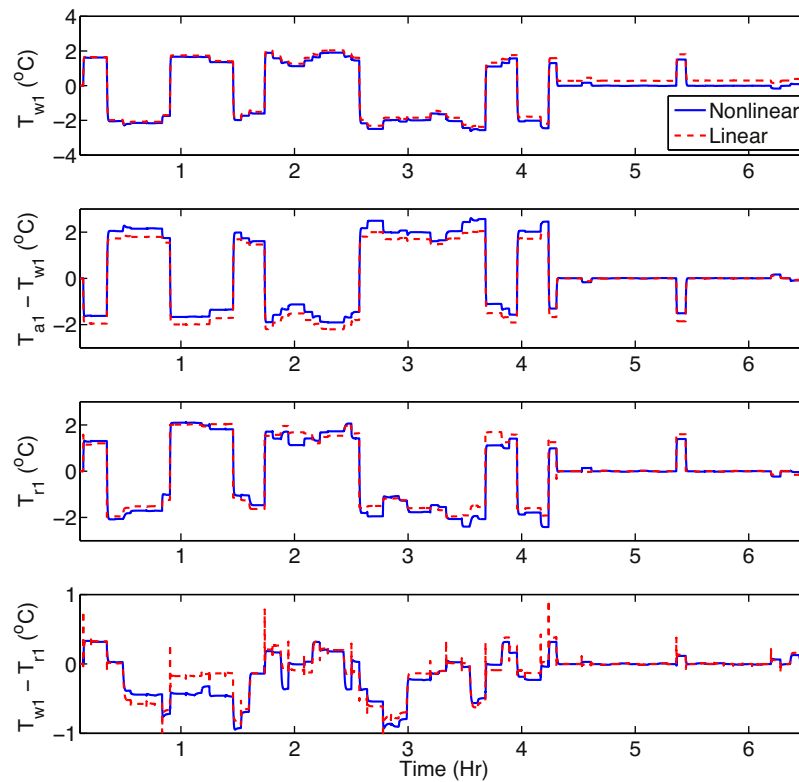


Fig. 23. Parameter identification validation of the thermal model through a comparison of the ATTMO nonlinear model and the linearized thermal dynamics model presented in Section 2.

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